



RESEARCH PROGRESS REPORT

Fast Radio Bursts as Probes of the Cosmic Baryon Distribution

Submitted by:
Agam Baruah
BS-MS Student
IISER Pune

Under the supervision of:
Dr. Prateek Sharma
Professor, Department of Physics
Indian Institute of Science (IISc)
Bengaluru, India

Indian Institute of Science Education and Research (IISER), Pune
August 2026

Contents

1 Acknowledgments	2
2 Title of Research Project	2
3 Aim/Objectives	2
4 Introduction to Research Topic	2
5 Theoretical Framework of Topic	3
6 Profile of Organisation/Research Lab	6
7 Methodology Followed	6
7.1 Data Aggregation & Standardization Pipeline	6
7.2 Selection Cuts and Cleaning Filters	6
7.3 Galactic and Halo Modeling	7
7.4 Convolved Likelihood	8
7.5 Use of IllustrisTNG Simulations	8
7.6 Alternative Host Galaxy Modeling: The Gamma Distribution	9
7.7 MCMC Parameter Sampling and Prior Distributions	10
7.8 Galaxy Overdensity Cross-Correlation: Physical Motivation and Pipeline	10
8 Analysis and Interpretation of Research Project	12
8.1 IGM Fraction and Host DM Parameter Constraints	12
8.2 Instrument-Specific Selection Biases	12
8.3 Cosmological Parameter Estimation: $\Omega_b h_{70}$	13
8.4 Bayesian Selection Function Implementation [19]	16
8.5 Scaling Behavior and the Roadmap Toward $\sim 5\%$ Precision in f_{IGM}	17
8.6 Host DM Standard Deviation Sensitivity Analysis	17
8.7 Systematic Multi-Tier Foreground and Milky Way Halo Distribution Inference	20
8.8 Jackknife Resampling Test	22
8.9 Comparison with the YMW16 Galactic ISM Density Model	23
8.10 Cosmic Web Gas Mapping: SDSS and DESI Cross-Correlations	24
9 Conclusion and Suggestion of Research Project	26
10 Results Achieved	26

Abstract

Fast Radio Bursts (FRBs) are powerful cosmological probes of the ionized baryon content of the Universe. We present a Bayesian analysis of a master catalog of $N = 80$ well-localized FRBs, constructed by cross-matching the Connor et al. (2025) sample with supplementary catalogs from CRAFT/ASKAP [10], DSA-110 [12], and MeerTRAP [11]. After applying rigorous selection criteria (Galactic latitude $|b| \geq 20^\circ$, signal-to-noise ratio ≥ 11.2 , Milky Way DM fraction < 0.4 , and burst-averaging of known repeaters to mitigate selection bias), we retain $N = 71$ sources for cosmological inference. We model the total DM budget using a bivariate log-normal distribution for the diffuse IGM and intervening halo contributions, calibrated against IllustrisTNG simulations [3], convolved with a log-normal host galaxy model, and perform MCMC sampling to constrain the IGM baryon fraction f_{IGM} , the bound halo fraction f_{halo} , and host galaxy parameters. Our fiducial analysis yields $f_{\text{IGM}} = 0.671_{-0.10}^{+0.11}$ and $f_{\text{halo}} = 0.237_{-0.08}^{+0.09}$, consistent with the Macquart relation [4] and indicating that $\sim 67\%$ of cosmic baryons reside in the diffuse warm-hot intergalactic medium. Binned $\langle DM_{\text{cos}} \rangle$ fitting yields $\Omega_b h_{70} = 0.0505 \pm 0.0046$ with a fixed Milky Way halo term of 30 pc cm^{-3} (a 4% uncertainty reduction over baseline samples). Systematic robustness is validated via Galactic foreground model comparison (NE2001 vs. YMW16, $\Delta f_{\text{IGM}} \approx 0.013$), alternative host distribution models (gamma with selection-biased decaying prior, yielding $\Omega_b h_{70} = 0.0490 \pm 0.0044$), and a host DM evolution analysis ($\alpha = 0.89_{-1.23}^{+0.91}$, consistent with zero evolution). Telescope-resolved analysis reveals significant inter-instrument tension in host DM parameters, identifying selection bias as the dominant systematic for future joint analyses. We additionally report a model-independent detection of large-scale cosmic structure in FRB dispersion measures via partial Spearman correlation with SDSS DR18 ($\rho_{\text{partial}} = 0.730, p = 0.0009$) and DESI DR9 ($\rho_{\text{partial}} = 0.469, p = 0.0030$) galaxy overdensities ($\sim 3\sigma$ detection), corroborating the baryon-structure connection independently of MCMC assumptions.

1 Acknowledgments

I would like to express my sincere gratitude and deep appreciation to my project supervisor, **Dr. Prateek Sharma** (Professor, Department of Physics, Indian Institute of Science, Bengaluru), for his constant guidance, support, and mentorship throughout this research project. I began this project with no prior background in FRB cosmology. I am particularly grateful for his patience in explaining concepts in detail, no matter how elementary my questions were. His insights and feedback have been central in shaping my understanding and approach to this study.

I am highly grateful to the **Indian Academy of Sciences (IASc), Bengaluru**, for selecting me for this project under the Summer Research Fellowship Program (SRFP). I appreciate their support in providing accommodation and meals during my tenure, which allowed me to focus fully on my research. I also extend my thanks to the Academy for organizing several enriching outreach programs, which were highly informative and beneficial to my academic development.

I am also thankful to the faculty, staff, and administration of the **Indian Institute of Science, Bengaluru**, for hosting me and providing access to the institute's academic facilities, library resources, and high-speed internet. I also benefited greatly from the institute's research presentation sessions, where I had the opportunity to present my work and receive invaluable feedback from faculty and peers, which helped me understand the project in a much deeper and more rigorous manner. Furthermore, I thank my home institution, **IISER Pune**, for its support and academic environment, which enabled me to undertake this work.

Finally, I acknowledge the **Department of Science and Technology (DST), Government of India**, for the financial assistance provided under

the Scholarship for Higher Education (SHE) component of the INSPIRE scheme, which has made this research experience possible.

2 Title of Research Project

Fast Radio Bursts as Probes of the Cosmic Baryon Distribution.

3 Aim/Objectives

The main objectives of this research project are as follows:

1. Compile and process the Canadian Hydrogen Intensity Mapping Experiment (CHIME)/FRBs Catalog 1, FRBCAT, and TNS databases into a unified, filtered pipeline of localized FRBs with known redshifts.
2. Compute and compare the extragalactic dispersion measures (DM_{IGM}) against the theoretical expectation derived from the cosmological baryon density.
3. Fit for the diffuse intergalactic baryon fraction (f_{IGM}) using convolved likelihoods and characterize the dispersion scatter around the relation.
4. Formulate and implement alternative host galaxy dispersion measure models (such as the Gamma distribution) with a selection-biased decaying prior on the shape parameter, and compare their cosmological parameter constraints and outlier mitigation properties to the fiducial log-normal model.
5. Investigate whether the rest-frame host galaxy dispersion measure evolves with redshift using a 5-parameter MCMC framework.
6. Perform instrument-resolved MCMC analyses on localized FRB subsets (DSA-110, ASKAP, MeerTRAP) to quantify population selection effects and systematic parameter shifts.
7. Cross-correlate FRB dispersion measures with galaxy over-density maps along the same sightlines to map cosmic web gas, using the SDSS and DESI Legacy Surveys.

4 Introduction to Research Topic

Standard cosmological models (derived from Big Bang Nucleosynthesis, BBN, and the Cosmic Microwave Background, CMB) predict that baryonic matter constitutes approximately 4.9% of the Universe's total energy density ($\Omega_b \approx 0.049$). However, direct observations of stars, interstellar dust, and cold gas in the local Universe ($z < 2$) fail to account for nearly half of these expected baryons (≈ 0.021 [1]). This observed discrepancy is known as the **missing baryons problem**.

Hydrodynamical simulations indicate that these "missing baryons" reside in a diffuse, highly ionized gas phase distributed along cosmic web filaments, known as the Warm-Hot Intergalactic Medium (WHIM) [2, 3]. Because of its high temperature ($T \sim 10^5 - 10^7 \text{ K}$) and extremely low density, the WHIM has remained largely invisible to traditional absorption and emission spectroscopy, as its emission is extremely faint.

Because FRBs are faint and travel through both Galactic and extragalactic plasma, high time resolution and high-latitude observations are essential to isolate extragalactic dispersion measures from Galactic interstellar medium contamination.

Fast Radio Bursts (FRBs) are extremely bright, millisecond-duration pulses of coherent radio emission originating at cosmological distances, providing a novel solution to this problem [4]. Although FRBs remain fundamentally unsolved at the source level, magnetars are currently the leading progenitor candidates. As an FRB pulse propagates through ionized plasma, its group velocity varies with frequency, causing lower-frequency radio waves to arrive later than higher-frequency waves. The magnitude of

Cosmic Baryon Accounting with Fast Radio Bursts: Multi-Survey Bayesian Framework

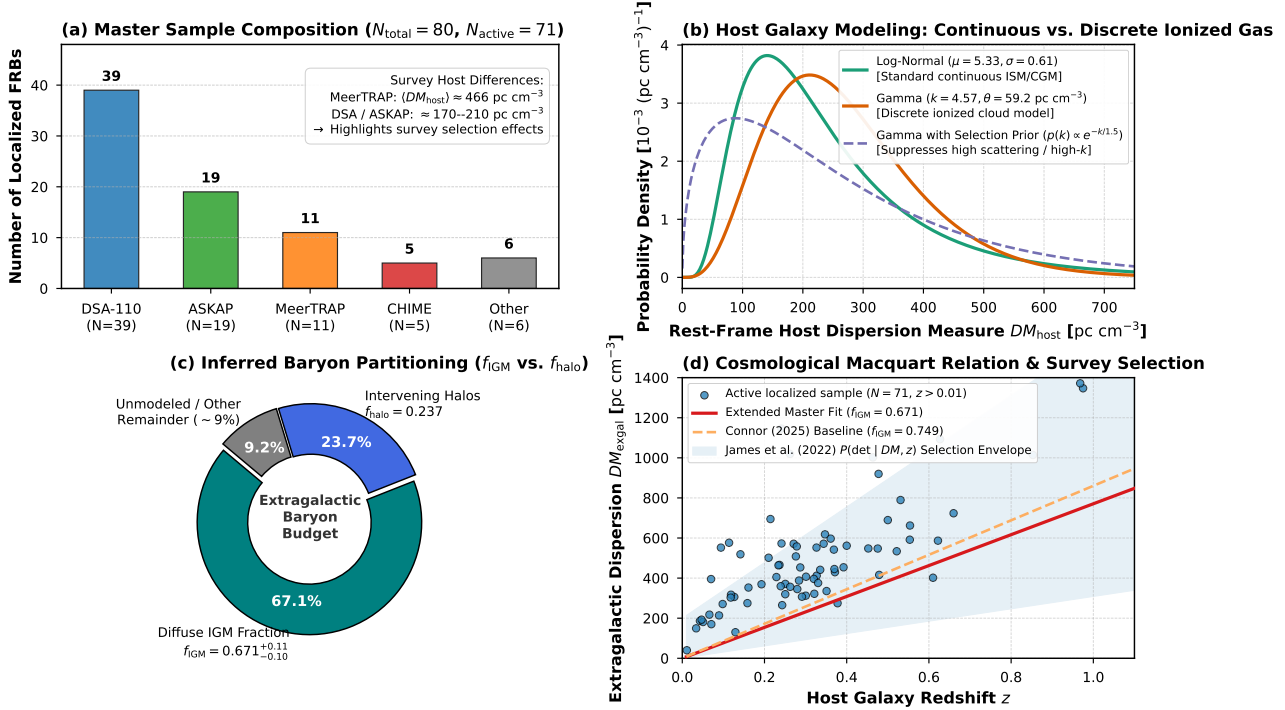


Figure 1: Graphical Abstract: Comprehensive multi-survey Bayesian framework for Fast Radio Burst cosmology. (a) Master sample composition ($N_{\text{total}} = 80$, $N_{\text{active}} = 71$) highlighting inter-instrument host DM differences. (b) Dual host galaxy modeling comparing continuous log-normal ISM/CGM distributions with a discrete ionized clump Gamma distribution regularized by a selection-biased decaying prior ($p(k) \propto e^{-k/1.5}$). (c) Constrained extragalactic baryon budget showing the diffuse IGM ($f_{\text{IGM}} = 0.671^{+0.11}_{-0.10}$) and intervening halo ($f_{\text{halo}} = 0.237$) fractions. (d) Cosmological Macquart relation for the active localized sample ($N = 71$) with the extended fit, baseline comparison, and observational selection envelope.

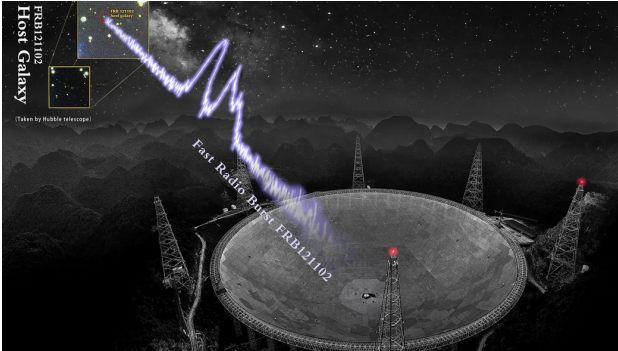


Figure 2: Artist's impression of FAST observing a pulse from FRB 121102. (Image Credit : NAOC)

this delay is directly proportional to the integrated free-electron column density (n_e) along the sightline, known as the **Dispersion Measure** (DM).

$$DM = \int_0^d n_e dl \quad (4.1)$$

$$\Delta t \text{ (time delay)} = \frac{e^2}{2\pi c m_e} \times DM \times \left(\frac{1}{v_1^2} - \frac{1}{v_2^2} \right) \quad (4.2)$$

It is important to distinguish dispersion from **scattering**, which is a separate and physically distinct propagation effect also caused by free electrons in the intervening plasma. While dispersion arises from the frequency-dependent group velocity of radio waves in a cold plasma, causing a smooth, deterministic v^{-2} arrival-time sweep across the band, scattering results from multipath propagation driven by density fluctuations and turbulent irregularities in the electron distribution. Scattered radio waves

travel slightly different effective path lengths before reaching the observer, causing the pulse to arrive smeared over a range of times and producing a characteristic one-sided exponential tail in the pulse profile (temporal broadening), with the broadening timescale scaling steeply as $\tau_{\text{scat}} \propto \nu^{-4}$. Crucially, dispersion is fully reversible: applying the known ν^{-2} delay correction (dedispersion) exactly recovers the intrinsic pulse shape. Scattering, however, is irreversible, the scattered pulse generally cannot be perfectly deconvolved without a precise model of the scattering screen. For this reason, DM is the cosmologically useful observable, encoding the total free-electron column density along the sightline, while the scattering timescale τ_{scat} probes plasma turbulence structure and must be carefully disentangled in high- DM sources where pulse broadening can mask the true temporal profile.

Because FRBs propagate across vast cosmological distances, their dispersion measures act as direct probes of the intergalactic medium and offer a unique method to locate and weigh the missing cosmic baryons [5].

5 Theoretical Framework of Topic

The observed dispersion measure of a cosmological FRB along a specific line of sight is modeled as a linear sum of contributing plasma environments:

$$DM_{\text{obs}} = DM_{\text{MW,ISM}} + DM_{\text{MW,halo}} + DM_{\text{IGM}}(z) + \frac{DM_{\text{host}}}{1+z} + \sum_i^{N_X} \frac{DM_X(M_i, b_{\perp})}{1+z_i} \quad (5.1)$$

where each term corresponds to a distinct physical component along the propagation path:

- DM_{obs} is the total observed dispersion measure.
- $DM_{\text{MW,ISM}}$ is the dispersion measure contribution from the Milky Way's interstellar medium (ISM), modeled using the NE2001 [6] or YMW16 [7] Galactic electron density models (see §7.3).

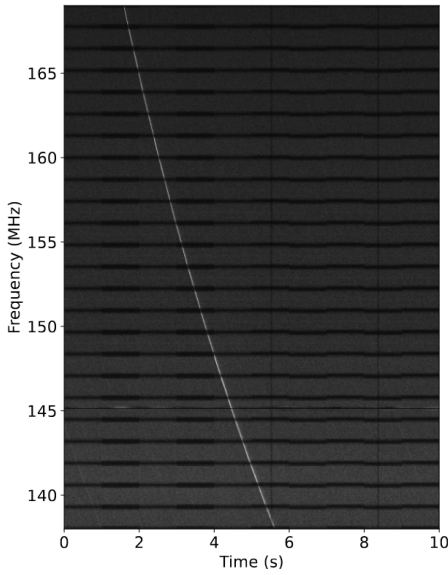


Figure 3: Dynamic spectrum showing the observed signal intensity (of a giant pulse from the Crab pulsar) as a function of frequency and arrival time (Bassa et al. 2017).

- $DM_{\text{MW,halo}}$ is the contribution from the Milky Way's extended circumgalactic medium (CGM), subtracted as a fixed constant (see §7.3).
- $DM_{\text{IGM}}(z)$ is the dispersion measure accumulated during propagation through the ionized intergalactic medium (IGM).
- $\frac{DM_{\text{host}}}{1+z}$ is the contribution from the host galaxy and its local environment. The factor of $\frac{1}{1+z}$ redshifts the local host-frame dispersion measure to the observer-frame, accounting for the cosmological expansion.
- $DM_X(M_i, b_\perp)$ is the rest-frame contribution from the i^{th} intersected halo of the N_X intervening halos.

Also, we define

$$DM_{\text{halo}} = \sum_i^{N_X} \frac{DM_X(M_i, b_\perp)}{1+z_i} \quad (5.2)$$

$$DM_{\text{cosmic}} = DM_{\text{IGM}} + DM_{\text{halo}} \quad (5.3)$$

$$DM_{\text{MW}} = DM_{\text{MW,halo}} + DM_{\text{MW,ISM}} \quad (5.4)$$

$$DM_{\text{ex}} = DM_{\text{cosmic}} + \frac{DM_{\text{host}}}{1+z} = DM_{\text{obs}} - DM_{\text{MW}} \quad (5.5)$$

The intergalactic medium contribution (DM_{IGM}) accumulates through propagation via the ionized cosmic web. Its redshift-dependent expectation value is described by the **Macquart relation**[4]:

$$\langle DM_{\text{IGM}}(z) \rangle = \frac{3cH_0\Omega_b}{8\pi Gm_p} \times \int_0^z \frac{f_d(z')f_{\text{IGM}}(z')(1+z')}{\sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}} dz' \quad (5.6)$$

Derivation of the Macquart Relation

The cosmological *scale factor* $a(t)$ describes the relative expansion of the Universe at cosmic time t , normalised to unity today: $a_0 \equiv$

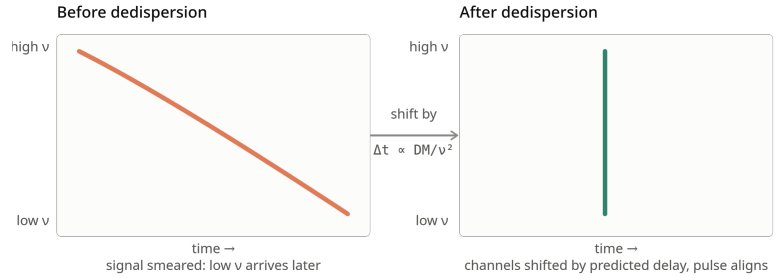


Figure 4: Schematic representation showing dispersed dynamic spectrum before and after dedispersion.

$a(t_0) = 1$. The cosmological redshift is defined by:

$$1+z = \frac{a_0}{a(t)} = \frac{1}{a(t)} \quad (5.7)$$

Differentiating both sides with respect to t , and using the Hubble parameter $H(z) \equiv \dot{a}/a$:

$$dz = -\frac{\dot{a}}{a^2} dt = -\frac{H(z)}{a} dt = -(1+z)H(z) dt \quad (5.8)$$

$$\Rightarrow dt = -\frac{dz}{(1+z)H(z)}$$

For a photon propagating along a null geodesic in the flat FLRW^a metric ($ds^2 = 0$), the proper path length element traversed in proper time $|dt|$ is:

$$dl = c|dt| = \frac{cdz}{(1+z)H(z)} \quad (5.9)$$

Because the Universe expands homogeneously, the comoving baryon number density is conserved. The *physical* baryon number density at redshift z therefore scales as:

$$n_b(z) = n_{b,0}(1+z)^3 \quad (5.10)$$

where $n_{b,0}$ is today's mean baryon number density. Not all baryons contribute free electrons to the IGM: a fraction $f_{\text{IGM}}(z)$ of cosmic baryons reside in the diffuse IGM, and of those, a fraction $f_d(z)$ are ionized. The physical free-electron number density is therefore:

$$n_e(z) = f_{\text{IGM}}(z)f_d(z)n_{b,0}(1+z)^3 \quad (5.11)$$

The Friedmann equation at $z = 0$, for a flat Universe, gives the critical density today:

$$\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} \quad (5.12)$$

The mean baryon mass density today is $\rho_{b,0} = \Omega_b \rho_{\text{crit},0}$. Approximating the mean baryon mass by the proton mass m_p :

$$n_{b,0} = \frac{\rho_{b,0}}{m_p} = \frac{\Omega_b \rho_{\text{crit},0}}{m_p} = \frac{3H_0^2 \Omega_b}{8\pi G m_p} \quad (5.13)$$

DM is an observer-frame quantity, inferred from $\Delta t_{\text{obs}} \propto v_{\text{obs}}^{-2} DM_{\text{obs}}$. Consider a thin plasma slab at redshift z' with proper depth dl' . In the rest-frame of that slab, the photon frequency is $v' = v_{\text{obs}}(1+z')$, so the local dispersion delay produced by that slab is:

$$\delta t' \propto \frac{n_e(z') dl'}{v'^2} = \frac{n_e(z') dl'}{v_{\text{obs}}^2 (1+z')^2} \quad (5.14)$$

Cosmological time dilation then stretches this delay by $(1+z')$ when received by the observer:

$$\delta t_{\text{obs}} = (1+z') \delta t' \propto \frac{n_e(z') dl'}{v_{\text{obs}}^2 (1+z')} = \frac{1}{v_{\text{obs}}^2} \times \frac{n_e(z') dl'}{(1+z')} \quad (5.15)$$

We define :

$$dDM_{\text{obs}} = \frac{n_e(z')}{1+z'} dl' \quad (5.16)$$

Substituting the results into $DM_{\text{IGM}} = \int dDM_{\text{obs}}$:

$$\begin{aligned} DM_{\text{IGM}}(z) &= \int_0^z \frac{n_e(z')}{1+z'} \cdot \frac{c}{(1+z') H(z')} dz' \\ &= c n_{b,0} \int_0^z \frac{f_{\text{IGM}}(z') f_d(z') (1+z')^3}{(1+z')^2 H(z')} dz' \\ &= c n_{b,0} \int_0^z \frac{f_{\text{IGM}}(z') f_d(z') (1+z')}{H(z')} dz' \end{aligned} \quad (5.17)$$

Using the flat- Λ CDM Hubble parameter $H(z') = H_0 \sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}$ and substituting $n_{b,0}$, we obtain the final expression of the Macquart relation:

$$\begin{aligned} \langle DM_{\text{IGM}}(z) \rangle &= \frac{3cH_0\Omega_b}{8\pi G m_p} \\ &\times \int_0^z \frac{f_{\text{IGM}}(z') f_d(z') (1+z')}{\sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}} dz' \end{aligned} \quad (5.18)$$

^aFriedmann–Lemaître–Robertson–Walker (FLRW) metric

where the cosmological and physical terms are defined as follows:

- c is the speed of light in a vacuum.
- H_0 is the Hubble constant, representing the current expansion rate of the Universe.
- Ω_b is the cosmological baryon density parameter.
- G is the gravitational constant.
- m_p is the proton mass.
- $f_d(z')$ is the free electron fraction of the intergalactic gas, predominantly consisting of Hydrogen and Helium gas. It is defined analytically as [8]:

$$f_d(z) = \frac{3}{4} y_1 \chi_{e,H}(z) + \frac{1}{8} y_2 \chi_{e,He}(z) \quad (5.19)$$

where $y_1 \approx 1$ and $y_2 \approx 1$ are normalization constants for the primordial composition (75% Hydrogen and 25% Helium by mass), and $\chi_{e,H}(z)$, $\chi_{e,He}(z)$ represent the ionization fractions of Hydrogen and Helium. Under the physical assumption of a fully ionized intergalactic medium at low redshifts ($z < 2$), this fraction evaluates to:

$$f_d = \frac{3}{4}(1) + \frac{1}{8}(1) = 0.875 \quad (5.20)$$

If the assumption of complete ionization is violated, the free electron fraction will drop, leading to an underestimation of the inferred cosmic baryon density (Ω_b).

- $f_{\text{IGM}}(z')$ is the fraction of cosmic baryons residing in the diffuse IGM.

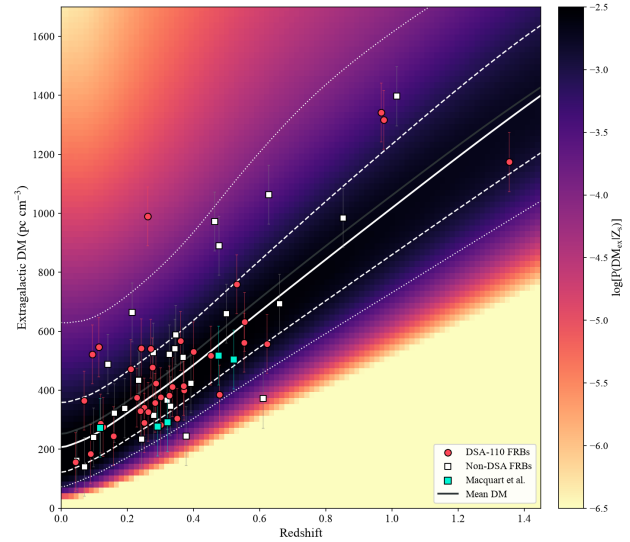


Figure 5: Macquart relation. DSA-110 (red circles), Non-DSA (white squares), and the original Macquart gold sample (cyan squares). The background shading shows the convolved 2D likelihood $\log_{10}[P(DM_{\text{ex}}|z)]$. The solid white line is the median, the dashed white lines are the 1σ contours, the dotted white lines are the 2σ contours, and the solid dark line is the mean $\langle DM_{\text{IGM}} \rangle(z)$ curve. The outlier FRB 20190520B is excluded.

- Ω_m and Ω_Λ are the cosmological density parameters for matter and dark energy, respectively. Here, $\Omega_\Lambda = 0.685$ and $\Omega_m = 0.315$.

This suggests that the diffuse intergalactic gas accounts for the primary redshift scaling of dispersion measures, while sightline scatter is dominated by stochastic host galaxy contributions.

Because the local plasma environment of the host galaxy varies, the host contribution (DM_{host}) is modeled as a stochastic variable following a log-normal probability density function[5]:

$$\begin{aligned} P(DM_{\text{host}}|\mu_{\text{host}}, \sigma_{\text{host}}) &= \frac{1}{DM_{\text{host}} \sigma_{\text{host}} \sqrt{2\pi}} \\ &\times \exp\left(-\frac{(\ln DM_{\text{host}} - \mu_{\text{host}})^2}{2\sigma_{\text{host}}^2}\right) \end{aligned} \quad (5.21)$$

where $e^{\mu_{\text{host}}}$ represents the median host dispersion measure, and σ_{host} represents the log-normal dispersion width. A log-normal distribution is appropriate because:

- DM_{host} is positive definite.
- Large values are very rare, so the distribution has a long positive tail.

To infer cosmological and host parameters, the joint likelihood $\mathcal{L}_i$ for each FRB i is calculated by convolving the intergalactic probability density $P(DM_{\text{IGM}}|z)$ (modeled from IllustrisTNG 300-1 simulation sightlines [3]) with the host log-normal distribution:

$$\mathcal{L}_i = P(DM_{\text{obs}}|z_i) = P(DM_{\text{cosmic}} + DM_{\text{host}}|z_i) \quad (5.22)$$

so,

$$\begin{aligned} \mathcal{L}_i &= \int_0^{DM_{\text{obs},i}} P(DM_{\text{cosmic},i} - DM'|z_i) \\ &\times P(DM'| \mu_{\text{host}}, \sigma_{\text{host}}) dDM' \end{aligned} \quad (5.23)$$

where $DM_{\text{cosmic},i} = DM_{\text{obs}} - DM_{\text{MW,ISM}} - DM_{\text{MW,halo}}$. Thus,

$$\begin{aligned} \mathcal{L} &= \prod_i^N \int_0^{DM_{\text{cosmic},i}} P(DM_{\text{obs},i} - DM'|z_i) \\ &\times P(DM'| \mu_{\text{host}}, \sigma_{\text{host}}) dDM' \end{aligned} \quad (5.24)$$

6 Profile of Organisation/Research Lab

This project was conducted within the Astronomy and Astrophysics group led by Prof. Prateek Sharma, in the Department of Physics at the Indian Institute of Science (IISc), Bengaluru. The Department of Physics at Indian Institute of Science (IISc), Bengaluru, established in 1933 under Nobel Laureate Sir C.V. Raman, is a leading centre of physical science research in India, with active programmes spanning condensed matter, high-energy physics, astrophysics, and plasma physics. Within this environment, Prof. Sharma's group actively works on all aspects of the CGM, the diffuse medium around Milky-Way like galaxies that contains the majority of the halo baryons. They have used high-resolution idealized simulations, complementary to the cosmological galaxy formation simulations that generally do not have a high enough resolution to faithfully simulate the multiphase CGM[9].

7 Methodology Followed

7.1 Data Aggregation & Standardization Pipeline

To build a homogeneous dataset, we aggregated localized FRB measurements from the CRAFT [10], MeerTRAP [11], DSA-110 [12], and Connor et al. [5] catalogs, supplemented by public alerts on the Transient Name Server (TNS). Each catalog was standardized and then merged through the following sequential steps:

1. **Schema and column pruning:** Retained only key physical parameters (redshift, DM, coordinates, survey metadata); removed NaN rows in critical fields.
2. **Source ID normalization:** Stripped FRB/FRB prefixes; resolved suffix discrepancies across catalogs (e.g., CRAFT's 20180924B → TNS 20180924A); removed duplicates via redshift and angular-coordinate closeness filters.
3. **Coordinate standardization:** Converted all sexagesimal and Galactic coordinates to decimal degrees (RA, Dec J2000).
4. **Data type normalization:** Mapped heterogeneous repeater flags (Y/N, yes/no, sub-name designations) to a boolean `True/False`; stripped residual LaTeX and parsing artefacts.
5. **Fit-to-burst DM for CHIME:** Used `dm_fitb` (profile-model fit) rather than `bonsai_dm` (coarse online trigger), as the former accounts for sub-burst frequency-dependent structure and yields a higher-fidelity physical measurement [13].¹
6. **Source priority hierarchy:** When a source appeared in multiple catalogs, primary survey publications (CRAFT, MeerTRAP, DSA-110) took precedence over TNS to preserve direct spectroscopic measurements.
7. **MeerTRAP NE2001 back-filling:** Nine MeerTRAP sources [11] were absent from TNS, leaving $DM_{MW,ISM}$ as NaN; values were manually extracted from the publication and back-filled.
8. **Unique TNS/FRBCAT integration:** Cross-matched against TNS and legacy FRBCAT; identified 2 unique events not already in the primary catalogs (FRB 20230808F and FRB 20210405I) and added them.

7.2 Selection Cuts and Cleaning Filters

To isolate the cosmological dispersion measure and minimize selection effects, the following sequential filters were implemented:

¹<https://www.chime-frb.ca/catalog>

7.2.1 Galactic Foreground Cuts

Two complementary filters are applied to suppress Galactic contamination.

1. All sightlines close to the Galactic disk ($|b| \leq 20^\circ$) are removed to minimize ISM modelling errors from the NE2001 and YMW16 electron density models, which are least reliable at low latitudes.
2. Nearby sightlines where the Milky Way ISM contribution dominates the observed DM, defined as $f_{MW} = DM_{MW,ISM}/DM_{obs} \geq 0.4$, are also excluded (e.g., dropping FRB 20220319D), since the extragalactic signal is too small to meaningfully constrain cosmological parameters.

7.2.2 Signal-to-Noise Ratio Cut

The signal-to-noise ratio (SNR) is a critical diagnostic of detection quality [14]. Low-SNR FRB events indicate noisy, uncertain signals where fitting a mathematical profile model to the burst shape is unreliable and yields high systematic errors in the fit-to-burst dispersion measure (DM_{fitb}).

Prior to cleaning, the raw database has a mean dispersion measure of $\langle DM_{obs} \rangle = 674.77 \text{ pc cm}^{-3}$. Applying the Galactic latitude disk cut and baseline SNR cleaning reduces the mean dispersion measure to $\langle DM_{obs} \rangle = 656.54 \text{ pc cm}^{-3}$. This decrease is physically expected because high-DM (high-redshift) bursts are intrinsically more distant and undergo more geometric dilution and scattering, making them more likely to be detected with low SNRs.

To determine the optimal SNR threshold, we perform a sensitivity analysis comparing two data-driven candidate cutoffs:

- **SNR = 11.2:** At this cutoff, the standard error of the mean (SE), which measures the statistical precision of our sample mean, achieves its global minimum, maximizing our confidence in the mean relation. The standard deviation (SD) also reaches a local minimum. This threshold retains a large sample size. This represents a rigorous, data-driven choice rather than an arbitrary conventional round number, optimizing the trade-off between sample size and measurement confidence.
- **SNR = 15.8:** Although the standard deviation reaches its lowest point here, the standard error is higher and increasing, and the sample size drops significantly. Crucially, the mean DM decreases significantly at this threshold, reflecting the deletion of a large number of high-DM FRBs. Removing these events biases cosmological inferences toward low-DM (nearby) FRBs and discards the high-redshift population where the Macquart relation's cosmological signal is most diagnostic [4].

Based on these statistics, we adopt $SNR = 11.2$ as our final signal-to-noise ratio threshold, optimizing the statistical precision of our cosmological sample².

An important physical observation is that even after applying the signal-to-noise ratio threshold and Galactic latitude cuts, the overall distribution shape of the dispersion measures remains remarkably identical. As shown in Figure 7, both the pre-cut and post-cut populations exhibit the characteristic right-skewed shape with a well-defined peak around $300 - 500 \text{ pc cm}^{-3}$ and a long, thin tail extending to high dispersion measures, representing the physical rarity of extreme dispersion events. The primary difference is a slight leftward shift of the distribution's peak and mean (from $\langle DM_{obs} \rangle = 674.77$ to $656.54 \text{ pc cm}^{-3}$), indicating the successful subtraction of the Milky Way disk ISM contribution. This confirms that the extragalactic features dominate the overall dispersion signal, providing a clean probe for cosmological modeling. Having fixed the SNR threshold, we next address a second selection-bias source: the treatment of repeating FRB sources within the catalog.

²This test is not a substitute for Jackknife resampling test. It protects the data quality for the Jackknife test to be meaningful which assumes the data points to be individually valid and tests for the robustness.

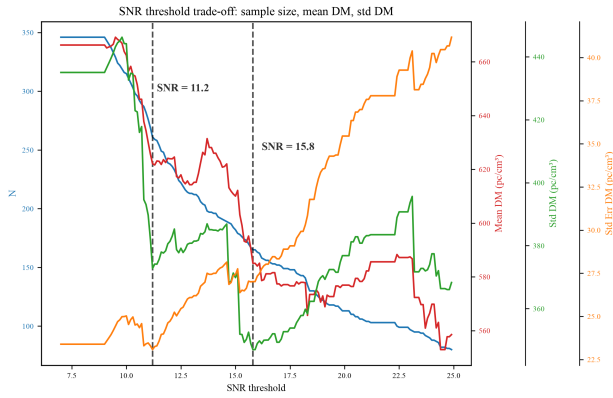


Figure 6: Sensitivity analysis of the FRB sample properties as a function of the signal-to-noise ratio (SNR) threshold. The plot shows the trade-off between the remaining sample size N (blue), mean DM (red), standard deviation of DM (green), and the standard error of DM (orange).

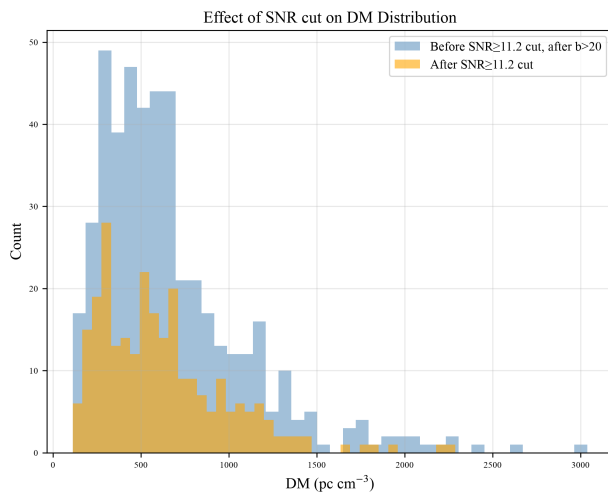


Figure 7: Dispersion measure (DM) distributions of the sample before and after the $SNR \geq 11.2$ selection cut. The overall right-skewed shape and high-DM tail remain identical, while the peak shifts slightly leftward due to Galactic foreground subtraction.

7.2.3 Repeater Handling and Selection Bias Mitigation

Initially, complete exclusion of repeating FRBs was considered. Physically, the plasma in the host environments of repeaters can evolve over time, meaning different bursts probe slightly different plasma states, and simple DM averaging would conflate these distinct physical measurements. Crucially, if we do not take care of repeating sources by grouping and averaging their bursts, the sample statistics become heavily biased toward the most active repeating sources (such as FRB20121102A or FRB 20190520B), which have emitted hundreds of detected bursts. Furthermore, repeaters may reside in physically different environments (such as younger, denser, or more highly magnetized circumgalactic structures), and mixing them directly without averaging or careful treatment would severely skew the DM distribution and bias the cosmological inferences. A boxplot comparison of the repeater and non-repeater populations in CHIME Catalog 1 [13]³ shows that repeaters are systematically clustered at lower DMs, with a median of $\sim 340 \text{ pc cm}^{-3}$ compared to $\sim 510 \text{ pc cm}^{-3}$ for non-repeater counterparts (a difference of $\sim 170 \text{ pc cm}^{-3}$). This is physically expected since lower-DM (closer) sources are easier to detect repeatedly. Excluding repeaters entirely would artificially skew the sample toward higher DMs. Thus, repeating sources are retained in the

³In the CHIME Catalog 1 database, repeaters are programmatically marked and flagged with a value of -9999 in the sub-burst and associated columns to distinguish them from single-burst events.

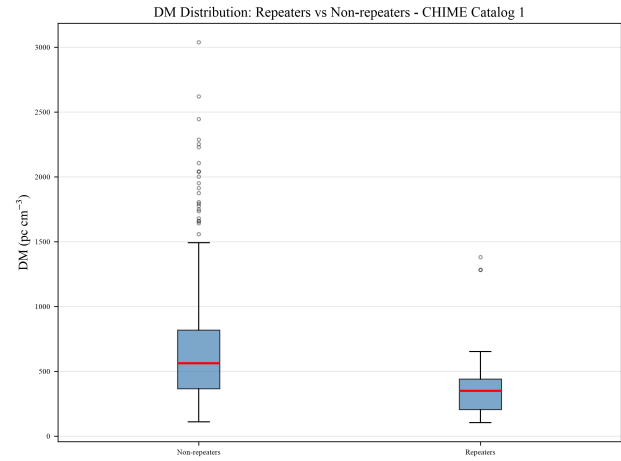


Figure 8: Dispersion measure (DM) distributions for repeaters and non-repeaters in the CHIME/FRB Catalog 1 [13], demonstrating the lower median DM of the repeating population ($\sim 340 \text{ pc cm}^{-3}$ vs. $\sim 510 \text{ pc cm}^{-3}$) and the statistical selection bias introduced if repeaters are excluded.

final sample by averaging their bursts, using their mean DMs to mitigate this selection bias. This is justified because the observed DM values for individual bursts of each repeating source are highly consistent with each other, showing very little variation.

7.2.4 Outlier Removal

We identified and surgically excised the anomalous source FRB20190520B ($z = 0.241$, $DM_{\text{ex}} \approx 1116.1 \text{ pc cm}^{-3}$). Its host contribution is estimated at $\approx 900 \text{ pc cm}^{-3}$ (nearly five times the cosmological expectation for its redshift), which would heavily skew the host log-normal parameter estimation and artificially suppress f_{IGM} .

The physical justification for this excision is strongly reinforced by recent observational and hydrodynamic modeling of FRB host environments. Follow-up optical spectroscopy of the host galaxy J160204.31–111717.3 [15] revealed it to be a low-mass dwarf galaxy ($\log_{10}(M_*/M_\odot) \approx 8.78$, $M_* \sim 6 \times 10^8 M_\odot$, approximately two orders of magnitude less massive than the Milky Way) with sub-solar gas-phase metallicity ($Z \approx 0.2\text{--}0.4 Z_\odot$, based on optical emission line ratios). As recently demonstrated by Diggins, Leung & Lu (2026) [16], young ($\lesssim 10 \text{ Myr}$), massive ($10^5\text{--}10^6 M_\odot$) stellar associations formed in low-metallicity, dust-poor dwarf galaxies lack the dust grains required to absorb Lyman-continuum photons. The unhindered ionizing flux generates giant, dense H II regions that can sustain local electron columns exceeding $DM_{\text{HII}} \gtrsim 10^3 \text{ pc cm}^{-3}$ along embedded sightlines. The extreme excess observed in FRB 20190520B is therefore an astrophysical property of a dust-poor, compact H II birth environment rather than a failure of the cosmological Macquart relation. Treating this source as an astrophysical outlier is thus physically well-founded, preventing its extreme local column from artificially inflating the host dispersion width or distorting the diffuse IGM budget. The final cleaned catalog contains $N = 80$ eligible localized FRBs spanning a redshift range of $0.0008 \leq z \leq 1.354$.

7.3 Galactic and Halo Modeling

To retrieve the true cosmological and host dispersion signals, we must isolate the extragalactic dispersion measure DM_{ex} from the observed dispersion DM_{obs} . For modeling the Galactic disk ISM contribution ($DM_{\text{MW,ISM}}$), two primary electron density models are compared:

- **NE2001 [6]:** Built using pulsar dispersion and scattering data, this model has been the standard baseline in cosmological analyses.
- **YMW16 [7]:** A more recent alternative utilizing a larger pulsar dataset and updated spiral arm structure geometries.

A comparison of the excess dispersion measures derived from these two

Galactic models shows that the values lie almost exactly along the 1 : 1 line, demonstrating strong agreement for the majority of high-Galactic latitude FRBs in our sample in Figure 10. The mean difference between the models is small:

$$\langle DM_{\text{exc,NE2001}} - DM_{\text{exc,YMW16}} \rangle \approx 4.71 \text{ pc cm}^{-3} \quad (7.1)$$

However, noticeable scatter away from the 1 : 1 line is observed. This modeling systematic occurs primarily near the Galactic plane ($|b| < 20^\circ$) or within dense Galactic regions, where clumpy local plasma features make the electron density profile extremely difficult to model.

Furthermore, neither the NE2001 nor the YMW16 model includes the diffuse gas residing in the extended Galactic halo. Manually subtracting the circumgalactic halo contribution is therefore required. While several models of the Galactic halo suggest a gas contribution of $DM_{\text{MW,halo}} \approx 50 \text{ pc cm}^{-3}$, we adopt a standardized value of:

$$DM_{\text{MW,halo}} = 30 \text{ pc cm}^{-3} \quad (7.2)$$

to maintain consistency with the baseline conventions of Connor et al. 2025 [5]. This subtraction is modeled as a constant, isotropic contribution because high-Galactic latitude sightlines exhibit negligible angular variation in Galactic halo models, and the outer halo's properties are poorly constrained compared to the disk, making a fixed subtraction a robust first-order approximation. Crucially, since $DM_{\text{MW,halo}}$ enters the dispersion measure partitioning as an additive constant, any residual deviation in the true Galactic halo contribution is absorbed directly into the inferred rest-frame host galaxy contribution.

7.4 Convolved Likelihood

Following the DM budget partitioning defined in Equation 5.1 and the Galactic foreground subtraction described in §7.3, assuming that the host galaxy contribution DM_{host} is physically independent of the cosmic pair (DM_{cosmic}), the PDF of their sum is given by the mathematical convolution of their individual PDFs:

$$\begin{aligned} P(DM_{\text{ex}} | z, \theta) &= (P_{\text{cosmic}} * P_{\text{host}})(DM_{\text{ex}} | z, \theta) \\ &= \int_0^{DM_{\text{ex}}} P_{\text{cosmic}}(DM_{\text{ex}} - DM' | z) \\ &\quad \times P_{\text{host}}(DM' | z, \mu_{\text{host}}, \sigma_{\text{host}}) dDM' \end{aligned} \quad (7.3)$$

The joint cosmic component is modeled as a bivariate, correlated log-normal distribution in the halo and IGM dispersion contributions:

$$\begin{aligned} P_{\text{cosmic}}(DM_{\text{halo}}, DM_{\text{IGM}} | z, \theta) &= \frac{A}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \\ &\quad \exp\left(-\frac{w_x^2 - 2\rho w_x w_y + w_y^2}{2(1-\rho^2)}\right) \\ &\quad \times \frac{1}{DM_{\text{halo}} \cdot DM_{\text{IGM}}} \end{aligned} \quad (7.4)$$

where $w_x = (\ln DM_{\text{halo}} - \mu_x)/\sigma_x$ and $w_y = (\ln DM_{\text{IGM}} - \mu_y)/\sigma_y$, and ρ is the correlation coefficient between the two components.

Computational Implementation: In its raw formulation, calculating the convolved likelihood requires a double numerical integration over the 2D cosmic grid at each MCMC step for every FRB:

$$\begin{aligned} P(DM_{\text{ex}} | z, \theta) &\propto \int_0^{DM_{\text{ex}}} dDM_{\text{IGM}} \int_0^{DM_{\text{ex}} - DM_{\text{IGM}}} \\ &\quad dDM_{\text{halo}} \times P_{\text{cosmic}}(DM_{\text{halo}}, DM_{\text{IGM}} | z) \\ &\quad \times P_{\text{host}}(DM_{\text{ex}} - DM_{\text{halo}} - DM_{\text{IGM}} | z) \end{aligned} \quad (7.5)$$

In the original code, this was evaluated as a $\mathcal{O}(n_{\text{FRB}} n_{\text{DM}}^3)$ 3D multiplication over a $(100, 100, 100)$ grid, allocating a 3D intermediate array of 1,000,000 floats 100 times per likelihood call, causing massive memory bandwidth and CPU bottlenecking.

To accelerate the solver, we implemented a discrete grid factorization based on a physical conservation insight: the host galaxy contribution $DM_{\text{host}} = (DM_{\text{ex}} - DM_{\text{cosmic}})(1+z)$ only depends on the combined sum of the cosmic components $DM_{\text{cosmic}} = DM_{\text{halo}} + DM_{\text{IGM}}$, rather than on their individual partitions. Because the grid resolutions for DM_{halo} and DM_{IGM} are identical linear arrays of size $N \times N$, lines of constant DM_{cosmic} are exactly the anti-diagonals of the 2D cosmic PDF grid.

We collapse the 2D grid $(100, 100)$ into a 1D array of unique cosmic sums D_s of size (199) ,⁴ by summing the cosmic probabilities along the anti-diagonals first:

$$\begin{aligned} P_{\text{cosmic, sum}}(D_s | z, \theta) &= \sum_{i+j=D_s} P_{\text{cosmic}}(DM_{\text{halo},i}, \\ &\quad DM_{\text{IGM},j} | z, \theta) \end{aligned} \quad (7.6)$$

We then construct a pre-computed 2D host galaxy probability matrix $P_{\text{host, 2D}}$ of shape $(100, 199)$ (observed extragalactic DM vs unique cosmic sums) and evaluate the convolved probability as a single matrix-vector product:

$$P(DM_{\text{ex}} | z, \theta) = P_{\text{host, 2D}} \cdot P_{\text{cosmic, sum}} \quad (7.7)$$

This anti-diagonal reduction reduces the operation to 100×199 steps, completely avoids $\mathcal{O}(N^3)$ memory allocations, and yields about a 22-fold CPU execution speedup, reducing the end-to-end 2000-step production run from 45 hours to ≈ 2 hours. To prevent log-posterior numerical crashes for low-DM sightlines, the lower grid integration limit is regularized to $DM_{\text{min}} = 1.0 \text{ pc cm}^{-3}$.

7.5 Use of IllustrisTNG Simulations

The IllustrisTNG hydrodynamical simulation (TNG300-1 run) [3] forms the basis of the joint cosmic distribution P_{cosmic} at each redshift z . The TNG300-1 box has a side length of $205 h^{-1} \text{ Mpc}$ comoving and tracks the hydrodynamical evolution of 2500^3 dark matter particles and 2500^3 gas cells.

Thousands of mock sightlines are integrated through the simulation box at redshift snapshots ($z = 0.1, 0.2, \dots, 5.0$). For each sightline, the electron density is partitioned into:

- **Diffuse IGM (DM_{IGM}):** Intergalactic plasma outside massive halos ($M_{\text{halo}} < 10^{12} M_\odot$ or outside R_{500}).
- **Collapsed Halos (DM_{halo}):** Ionized gas within the CGM of intervening halos ($M_{\text{halo}} \geq 10^{12} M_\odot$ or within R_{500}).

By fitting the joint $(DM_{\text{IGM}}, DM_{\text{halo}})$ distributions at each snapshot, the 6 bivariate log-normal parameters ($\mu_x, \mu_y, \sigma_x, \sigma_y, \rho, A$) are tabulated as a function of redshift and interpolated to each FRB's redshift z_i using 1D cubic splines during MCMC sampling.

To avoid re-running new simulations for every trial value of $(f_{\text{IGM}}, f_{\text{halo}})$, we rescale the log-means analytically relative to the baseline TNG values ($f_{\text{IGM,TNG}} = 0.827, f_{\text{halo,TNG}} = 0.138$) [5]⁴:

$$\mu_y(z) = \mu_{y,\text{TNG}}(z) + \ln\left(\frac{f_{\text{IGM}}}{f_{\text{IGM,TNG}}}\right) \quad (7.8)$$

$$\mu_x(z) = \mu_{x,\text{TNG}}(z) + \ln\left(\frac{f_{\text{halo}}}{f_{\text{halo,TNG}}}\right) \quad (7.9)$$

Physically, it just rescales the typical amount of gas in the IGM channel keeping the distribution invariant.

⁴This assumes a first-order approximation: the shape of the distribution (ρ, σ) and the covariance structure of density fluctuations remain fixed; only the normalization (log-mean) changes.

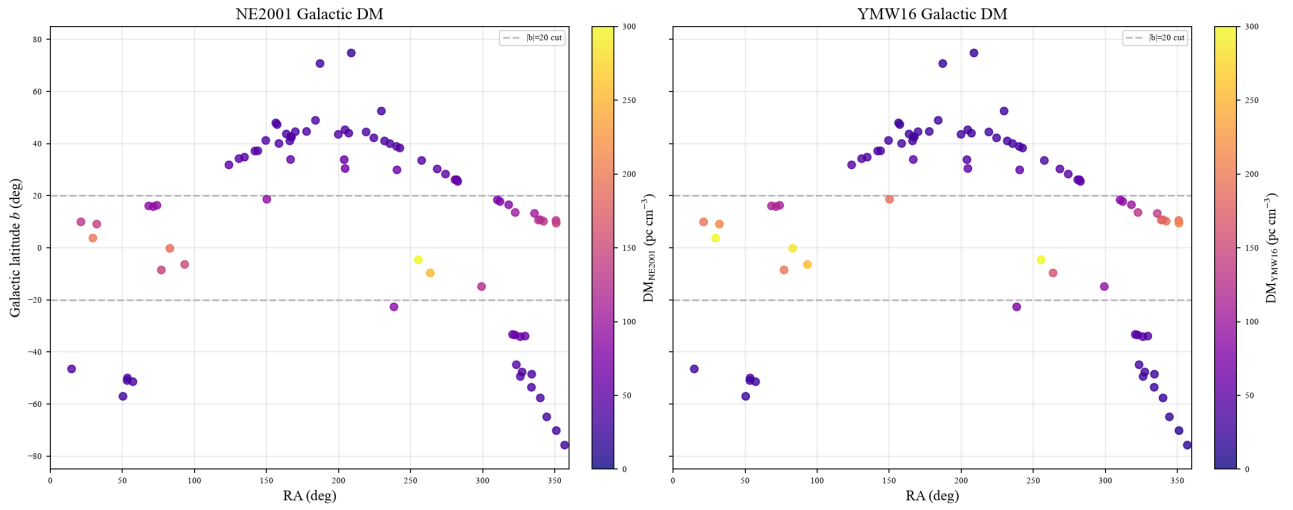


Figure 9: Comparison of the $DM_{MW,ISM}$ derived using the NE2001 and YMW16 Galactic disk ISM models for our localized sample.

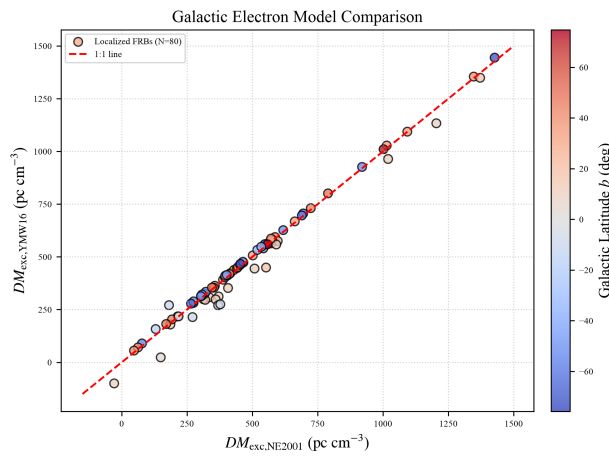


Figure 10: Comparison of the excess dispersion measures derived using the NE2001 and YMW16 Galactic disk ISM models for our localized sample. The dashed red line represents the 1:1 line of perfect agreement.

7.6 Alternative Host Galaxy Modeling: The Gamma Distribution

While the log-normal distribution is the standard choice in the primary literature [4, 5], we also implement and test a Gamma distribution model for the host galaxy DM.

Physically, the Gamma distribution is motivated by viewing the host interstellar medium (ISM) and circumgalactic medium (CGM) not as a smooth continuous fluid, but as a collection of discrete ionized clouds. If a line of sight through a host galaxy intersects a random, discrete number of ionized clouds, and each cloud contributes an independent and exponentially distributed dispersion measure.

$$\delta DM_i \sim \text{Exp}(\theta) \quad (7.10)$$

$$\text{Exp}(\theta) = \begin{cases} \theta e^{-\theta x}, & x \geq 0, \\ 0, & x < 0. \end{cases} \quad (7.11)$$

Then the total host contribution is the sum of these contributions:

$$DM_{\text{host}} = \sum_{i=1}^k \delta DM_i \sim \Gamma(k, \theta) \quad (7.12)$$

Mathematically, the sum of k independent and identically distributed exponential random variables follows an Erlang distribution⁵. Generalizing to non-integer values of the shape parameter k (which can phenomenologically represent geometric projection angles ϕ , clump size distributions, and partial line-of-sight edge clipping), the host galaxy DM in its rest-frame follows a Gamma distribution:

$$P_{\text{host}}(DM_{\text{host}} | k, \theta) = \frac{1}{\theta^k \Gamma(k)} (DM_{\text{host}})^{k-1} \times e^{-DM_{\text{host}}/\theta} \quad (7.13)$$

where $\Gamma(k)$ is the standard Gamma function.

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad (7.14)$$

Here, k is the shape parameter, and θ is the scale parameter, yielding a mean rest-frame host DM of $\langle DM_{\text{host}} \rangle = k\theta$.

This parameterisation introduces several physical and statistical advantages:

- **Flexibility at the Origin ($DM_{\text{host}} \rightarrow 0$):** A log-normal PDF is strictly pinned to zero at the origin ($P(DM_{\text{host}}) \rightarrow 0$ rapidly as $DM_{\text{host}} \rightarrow 0$). However, if an FRB is born in the clean outskirts of a host, in a globular cluster, or within a gas-poor dwarf galaxy, its host contribution can be negligible. A Gamma distribution with shape parameter $k \leq 1$ does not drop to zero at the origin; it can be finite or diverge, providing a natural mathematical model for clean, zero-host sightlines.
- **Lighter Tails and Outlier Mitigation:** The Gamma distribution exhibits an exponentially decaying tail (scaling as $e^{-DM_{\text{host}}/\theta}$), which is lighter than the sub-exponential, heavy tail of the log-normal distribution. This makes parameter estimation less sensitive to extreme local outliers (like FRB20190520B) which otherwise force a log-normal model to inflate its variance, biasing the entire population.

Furthermore, we account for observational selection effects in our priors. Real-world radio telescopes preferentially detect sharp, unscattered pulses, meaning our catalog is biased against FRBs passing through dense galactic disks (large k , high scattering) and biased toward clean sightlines (small k , low scattering). If we assume a flat uniform prior on k , the MCMC sampler will over-represent the heavily scattered, high- k population, leading to an

⁵Erlang is a special case of Gamma distribution restricted to integer shape parameter i.e., $E(k, \lambda) = \Gamma(k \in \mathbb{Z}^+, \lambda)$.

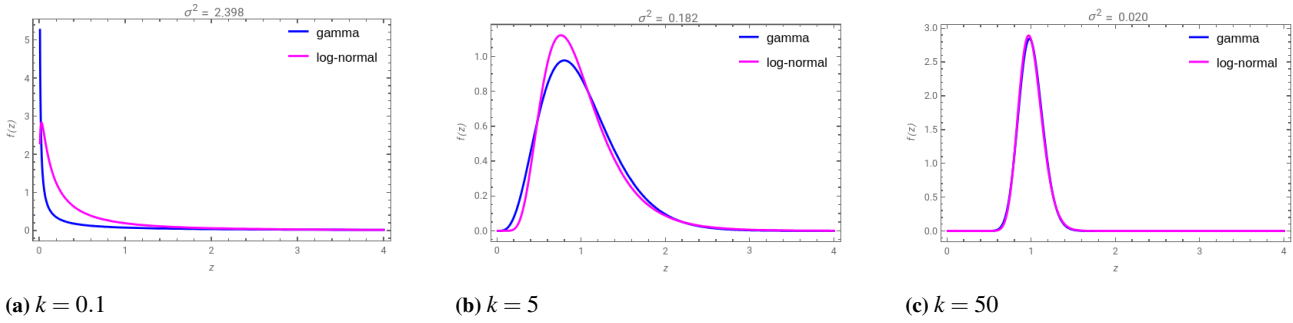


Figure 11: Comparison of the Gamma distribution $\Gamma(k, \theta = \frac{1}{k})$ and Log-normal distribution $L(\mu = 1, \sigma = \ln(1 + \frac{1}{k}))$.

overestimate of the average host DM. Because of the $\mu_{\text{host}}-f_{\text{IGM}}$ degeneracy, overestimating the host DM forces the MCMC to compensate by underestimating the cosmic slope, artificially depressing f_{IGM} . To correct for this, we implement a selection-biased decaying prior on the shape parameter:

$$p(k) \propto e^{-k/\lambda} \quad (7.15)$$

with decay scale $\lambda = 1.5^6$, which penalizes high- k configurations to reflect the telescope's selection function.

7.7 MCMC Parameter Sampling and Prior Distributions

Using the convolved probability $P(DM_{\text{ex}}|z, \theta)$, the joint likelihood for a catalog of N localized FRBs is:

$$\mathcal{L}(\theta) = \prod_{i=1}^N P(DM_{\text{ex},i}|z_i, \theta) \quad (7.16)$$

We sample the joint posterior distribution $p(\theta|\text{data}) \propto \mathcal{L}(\theta)p(\theta)$ using the affine-invariant ensemble MCMC sampler (`emcee`). The 4-parameter model fits the parameters $\theta = \{f_{\text{IGM}}, f_{\text{halo}}, \mu_{\text{host}}, \sigma_{\text{host}}\}$. The priors $p(\theta)$ are defined as [5]:

- Flat prior on $f_{\text{IGM}} \in [0, 1]$.
- Flat prior on $f_{\text{halo}} \in [0, 1]$.
- Flat prior on $\mu_{\text{host}} \in [0, 7.0]$.
- Flat prior on $\sigma_{\text{host}} \in [0.1, 2.5]$.
- A cosmological prior constraint: $f_{\text{IGM}} + f_{\text{halo}} < 1$, representing the physical upper limit of the diffuse cosmic baryon fraction.

The MCMC is initialized with 32 walkers. The sampler is run for 28000 iterations, discarding the first 1500 steps as burn-in. Walker convergence is verified by checking the autocorrelation time (τ), ensuring the production chains are run for multiple tens of τ to guarantee independent, uncorrelated posterior samples.

7.8 Galaxy Overdensity Cross-Correlation: Physical Motivation and Pipeline

When an FRB is observed with an exceptionally high dispersion measure (DM), a fundamental question arises: is a high DM more likely driven by a high cosmological redshift (z) or an intervening galaxy overdensity (δ_{gal})? At very high redshifts, the luminosity distance $d_L(z)$ is very large,

⁶The decay scale $\lambda = 1.5$ is chosen to center the prior expectation of the shape parameter k at 1.5, balancing the physical expectation that detectable, low-scattering sightlines intersect on average 1-2 ionized clumps in the host ISM, while preventing the sampler from entering the unphysical $k < 1$ regime where the Gamma probability density diverges at the origin, and penalizing high- k configurations that correspond to heavily scattered lines of sight.

meaning only the most intrinsically energetic, rare FRBs can exceed the survey detection threshold. Furthermore, even if such ultra-distant bursts are detected, their SNR is very unlikely to exceed stringent survey cutoffs due to pulse broadening and distance attenuation. Consequently, observed high- DM bursts are statistically far more likely to be caused by an extreme galaxy overdensity along a moderate-redshift sightline rather than an ultra-high redshift source [17]. This underscores why dispersion measure residuals (δ_{DM}) serve as a sensitive, direct probe of localized overdensities. The standard Macquart relation represents a statistical average over a homogeneous Universe. In the real Universe, matter is arranged in a cosmic web of nodes, filaments, and voids. An individual FRB sightline acts as a pencil beam traversing this structured medium:

- **Overdense sightlines:** Intersecting massive clusters or dense filaments, the sightline accumulates excess free electrons, yielding a positive dispersion residual: $\delta_{DM} = DM_{\text{cosmic}} - DM_{\text{Macquart}}(z) > 0$.
- **Underdense sightlines:** Paths through cosmic voids bypass the bulk of the cosmic gas, yielding $\delta_{DM} < 0$.

Galaxies trace the underlying dark matter and baryonic gas distribution, so the cumulative foreground galaxy overdensity $\Sigma\delta_{\text{gal}}$ serves as a normalized proxy for the total gas density along the sightline. Cross-correlating $\Sigma\delta_{\text{gal}}$ with δ_{DM} allows us to observationally verify that the scatter around the Macquart relation is driven by the cosmic web, providing a direct map of the Universe's diffuse baryons.

SDSS Cross-Correlation Pipeline. To search for direct observational evidence that FRBs trace the diffuse cosmological gas, we implemented a cross-correlation pipeline between FRB dispersion measures and foreground galaxy overdensities using Sloan Digital Sky Survey (SDSS) data. First, we mapped the sky footprint of the localized FRB sample against the SDSS coverage map, identifying $N_{\text{overlap}} = 18$ localized sources whose sightlines lie within the SDSS footprint (including 7 from DSA-110, 6 from ASKAP, 3 from CHIME, 1 from MeerTRAP, and 1 from Parkes). For each overlapping source, we queried the SDSS database (Data Release 18) for all PhotoObj/Photoz galaxy matches within a $0.3^\circ \times 0.3^\circ$ box centered on the FRB coordinates. To ensure the integrity of the intervening matter mapping, we cleaned the raw galaxy catalogs by applying a photo- z quality cut: we excluded galaxies with invalid or negative photo- z estimations ($z_{\text{photo}} \leq 0$), and discarded sources with a relative photo- z uncertainty exceeding:

$$\frac{z_{\text{photo, err}}}{z_{\text{photo}}} \geq 0.10 \quad (7.17)$$

Furthermore, we only retained galaxies satisfying $z_{\text{photo}} < z_{\text{FRB}}$ to isolate the foreground structures along the sightline. A simple total count of foreground galaxies along a sightline is heavily biased: more distant (high-redshift) FRBs naturally have longer paths and cross more galaxies (because of volume effect), and the sensitivity of the SDSS catalog drops sharply at higher redshifts due to the survey flux limit (selection function). To normalize for these effects, we implemented a shell-based cumulative overdensity method:

1. We query and clean galaxy catalogs for 100 random background fields

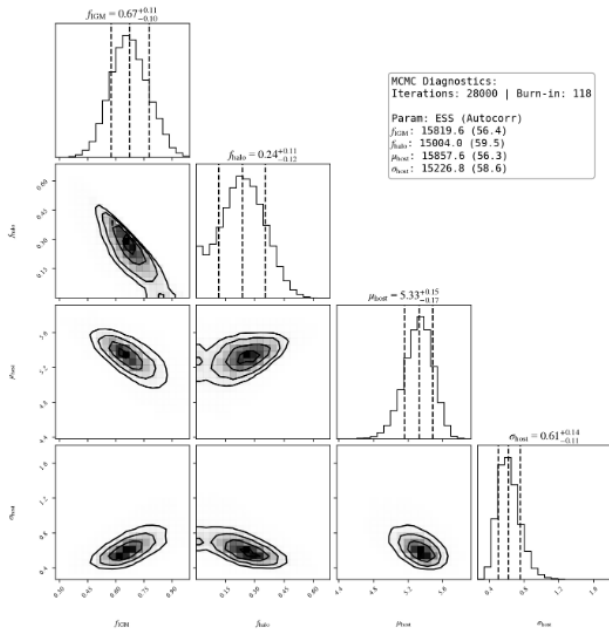


Figure 12: MCMC posterior corner plot showing the joint parameter constraints and correlations for the 4-parameter physical model (IGM fraction f_{IGM} , halo fraction f_{halo} , host galaxy median dispersion $e^{\mu_{\text{host}}}$, and log-normal width σ_{host}) evaluated on the extended catalog ($N = 80$).



Figure 14: Optical images of representative localized FRB sightlines with foreground galaxies retrieved from NED. The red cross denotes the FRB position, the dashed circle marks a projected physical radius of 2 Mpc, and cyan circles indicate foreground galaxies with $z < z_{\text{FRB}}$. The listed δDM values correspond to the FRB excess dispersion measures. This figure is intended to provide physical intuition for the connection between DM residuals and foreground galaxy overdensities; it is not based on the galaxy sample used in the statistical analysis. Owing to the incompleteness of NED, only relatively bright, previously catalogued galaxies are shown, and the true galaxy environment is significantly richer than depicted.

selected within the same SDSS footprint coordinates to establish a representative background density.

2. We divide the redshift space up to $z \approx 1.4$ into discrete shells of width $\Delta z = 0.05^7$.

⁷A redshift bin width of $\Delta z = 0.05$ is optimal for SDSS photo- z catalogs: it is slightly larger than the typical photo- z uncertainty ($\sigma_z \approx 0.02 - 0.03$), preventing galaxies from randomly scattering across multiple bins; it ensures the average expected count in the background field $\bar{n}$ remains statistically stable to avoid division-by-zero or excessive shot noise in the fractional overdensity $\delta_{\text{shell}} = (N_{\text{shell}} - \bar{n})/\bar{n}$; and it matches the physical scale of cosmic web structures ($\sim 140 h^{-1}$ Mpc radial thickness at $z \approx 0.3$) without averaging out voids and filaments.

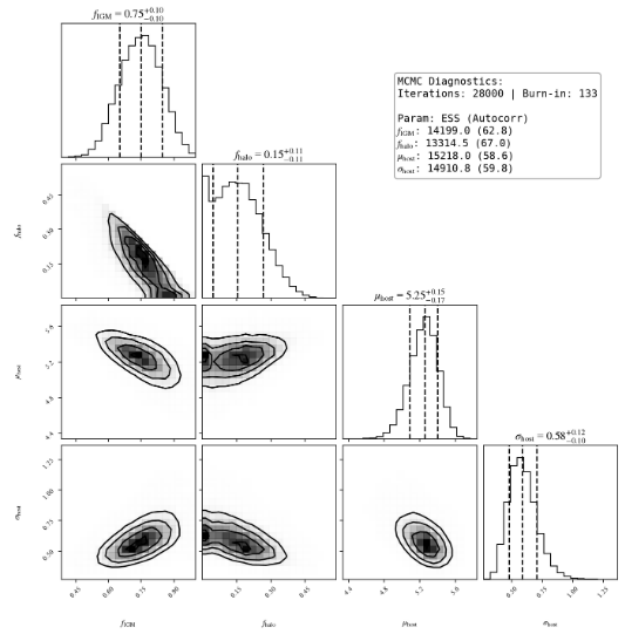


Figure 13: MCMC posterior corner plot showing the joint parameter constraints and correlations for the 4-parameter physical model (IGM fraction f_{IGM} , halo fraction f_{halo} , host galaxy median dispersion $e^{\mu_{\text{host}}}$, and log-normal width σ_{host}) evaluated on the Connor catalog ($N = 69$) for comparison purposes.

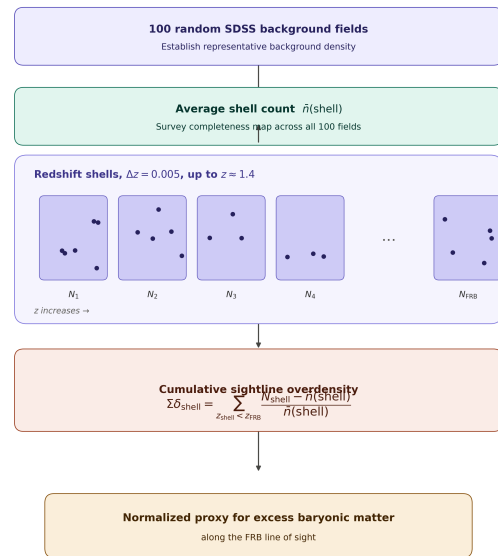


Figure 15: Flow diagram documenting the method for cross-correlation between dispersion measure residual and galaxy overdensity

3. We compute the average expected background galaxy count in each shell, $\bar{n}(\text{shell})$, across all 100 random fields, mapping the survey completeness function.
4. For each actual FRB sightline, the cumulative sightline overdensity $\Sigma \delta_{\text{shell}}$ is calculated by summing the shell-by-shell fractional overdensities up to the FRB redshift:

$$\Sigma \delta_{\text{shell}} = \sum_{z_{\text{shell}} < z_{\text{FRB}}} \frac{N_{\text{shell}} - \bar{n}(\text{shell})}{\bar{n}(\text{shell})} \quad (7.18)$$

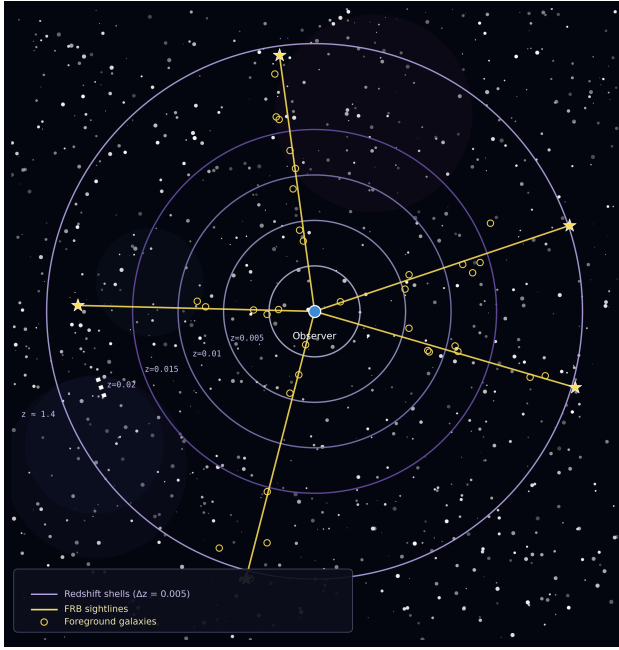


Figure 16: Schematic diagram to explain the redshift binning

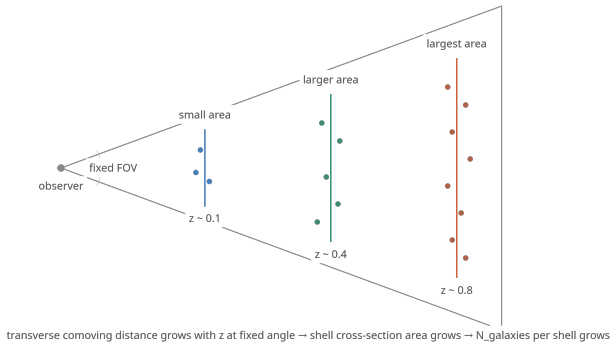


Figure 17: Schematic of survey cone geometry: a fixed angular aperture subtends a larger physical transverse area at higher redshift, increasing the shell volume and raw galaxy counts, which necessitates a shell-by-shell normalized overdensity calculation.

where N_{shell} is the count of foreground galaxies along the FRB sightline in that specific redshift bin. This cumulative overdensity serves as a normalized proxy for the total excess baryonic matter along the path.

DESI Legacy Survey Cross-Correlation Pipeline. To complement the SDSS result and verify signal robustness over a larger footprint with deeper photometry, we extended the same cross-correlation methodology to the DESI Legacy Imaging Survey (Data Release 9). The overall pipeline photo-z quality cuts ($z_{\text{phot, err}}/z_{\text{phot}} < 0.10$), foreground-only galaxy retention ($z_{\text{gal}} < z_{\text{FRB}}$), and shell-based cumulative overdensity calculation $\Sigma\delta_{\text{shell}}$, follows the SDSS procedure described above. The key differences are:

- **Larger overlap sample:** $N_{\text{overlap}} = 39$ sightlines (vs. 18 for SDSS), reflecting DESI's wider sky coverage.
- **Redshift source:** Spectroscopic z_{spec} from DESI/SDSS cross-matches where available; otherwise the random-forest photometric redshift median ($z_{\text{phot, median}}$) from the PRLS value-added catalog [18].
- **Local background calibration:** Rather than 100 global random fields, we generated 3 local background fields per sightline by per-

turbing the FRB coordinates by 1.2° – 2.2° , yielding 117 fields total. This accounts for DESI's spatially varying depth more robustly.

- **Queries:** Performed via the Astro Data Lab TAP service.⁸

8 Analysis and Interpretation of Research Project

8.1 IGM Fraction and Host DM Parameter Constraints

We executed the convolved MCMC on both the baseline catalog⁹ ($N = 69$) and the extended Master Catalog ($N = 80$)¹⁰. The posterior constraints are presented in Table 3.

The extended catalog yields $f_{\text{IGM}} = 0.67^{+0.11}_{-0.10}$, showing a decrease of ~ 0.08 compared to the baseline ($0.75^{+0.10}_{-0.10}$). This shift is interpreted as a physical selection effect: the newly added MeerTRAP and DSA-110 sources occupy systematically higher observed dispersion measures than the original Connor gold sample. To fit these high-DM sightlines within the same model, the convolved joint MCMC increases the preferred host galaxy contribution, shifting μ_{host} from 5.25 to 5.33 (corresponding to a median rest-frame host contribution of $e^{5.33} \approx 206.6 \text{ pc cm}^{-3}$ vs. $e^{5.25} \approx 190.6 \text{ pc cm}^{-3}$) and widening the host scatter from $\sigma_{\text{host}} = 0.58$ to 0.61. By the total DM budget equation:

$$DM_{\text{obs}} = DM_{\text{MW,ISM}} + DM_{\text{MW,halo}} + DM_{\text{IGM}} + \frac{DM_{\text{host}}}{1+z}, \quad (8.1)$$

allocating a larger portion of the extragalactic budget to the host galaxy naturally leaves a smaller residual for DM_{IGM} , which is directly proportional to f_{IGM} . This is the fundamental host-IGM degeneracy at the heart of all FRB-based baryon census measurements.

The choice of a log-normal distribution to model the host galaxy DM is physically well motivated. The host dispersion measure is the product of the electron number density n_e and the effective path length L through the host ISM and CGM: $DM_{\text{host}} \propto n_e L$. Both n_e and L are themselves distributed approximately log-normally across a heterogeneous galaxy population (reflecting the multiplicative stochastic processes that shape galactic star-formation histories, gas stripping events, and ISM turbulent pressure balance), so their product DM_{host} is also approximately log-normal by the multiplicative central limit theorem. The log-normal's long right tail further accommodates rare, extreme host environments such as FRB 20190520B (estimated host contribution $\approx 900 \text{ pc cm}^{-3}$), which would be catastrophically underrepresented by a symmetric Gaussian model.

The inferred f_{halo} and f_{IGM} are strongly anti-correlated within the joint posterior, as enforced by the physical baryon-conservation boundary $f_{\text{IGM}} + f_{\text{halo}} < 1$. This anti-correlation is directly visible as the elongated tilted posterior ellipses in the corner plots. The physical interpretation is that any portion of the extragalactic DM budget not attributed to the diffuse IGM must be partitioned among galactic halos (both the Milky Way circumgalactic medium and intervening foreground galaxy CGM halos), so the two fractions are causally coupled. Tighter priors on either parameter can break this degeneracy: observationally, this requires either independent per-source FRB host absorption measurements from high-resolution spectroscopy, or direct CGM constraints from UV absorption spectroscopy of intervening quasar sightlines, which motivates future multi-wavelength follow-up programs.

8.2 Instrument-Specific Selection Biases

We ran independent MCMCs for separate telescope instruments to investigate systematic differences. The preferred host galaxy and intergalactic fractions show significant tension¹¹:

⁸<https://datalab.noirlab.edu/tap>

⁹Baseline catalog refers to Connor catalog [5]

¹⁰Extended Master Catalog refers to the merged catalog from different sources[5, 10, 11, 12].

¹¹There is also 1 FRB from DSA-10 and 2 FRBs from MeerKAT.

Table 1: Localized FRBs overlapping the SDSS footprint with coordinates, galaxy query counts, DM residuals, and redshifts.

S.No.	FRB-id	RA (HMS)	Dec (DMS)	Query Galaxy Count	δ_{DM} (pc cm ⁻³)	Redshift
1	20180301A	06:12:54.432	04:40:15.960	4383	84.4268	0.3304
2	20180916B	01:58:00.744	65:43:00.480	3104	94.1384	0.0337
3	20190608B	22:16:04.776	-07:53:53.880	11154	183.0734	0.1178
4	20200120E	09:57:54.672	68:49:08.040	3964	16.5066	0.0008
5	20200430A	15:18:49.536	12:22:36.480	18034	199.0532	0.1610
6	20200906A	03:33:59.088	-14:04:59.520	6319	218.9156	0.3688
7	20201124A	05:08:03.504	26:03:38.520	9200	166.1095	0.0982
8	20210807D	19:56:53.136	-00:45:44.640	4650	1.1670	0.1293
9	20211212A	10:29:24.168	01:21:37.800	12576	86.8476	0.0707
10	20220105A	13:55:13.008	22:27:57.960	16650	309.4945	0.2785
11	20220307B	23:23:29.880	72:11:32.640	1315	145.0672	0.2507
12	20220319D	02:08:42.696	71:02:07.080	2010	1.7449	0.0111
13	20220506D	21:12:10.752	72:49:38.280	159	45.7585	0.3005
14	20221219A	17:10:31.152	71:37:36.480	4870	183.9462	0.5540
15	20230216A	10:25:53.328	03:26:12.120	15219	330.5350	0.5310
16	20230307A	11:51:07.584	71:41:42.360	4844	328.8247	0.2710
17	20230521B	23:24:08.640	71:08:16.800	573	63.3090	1.3540
18	20230907D	12:28:34.200	08:39:29.130	15567	599.6654	0.4638

- **MeerTRAP Subsample** ($N = 11$): $\mu_{\text{host}} = 6.14^{+0.44}_{-0.64}$, $f_{\text{IGM}} = 0.40^{+0.22}_{-0.23}$.
- **DSA-110 Subsample** ($N = 39$): $\mu_{\text{host}} = 5.35^{+0.18}_{-0.21}$, $f_{\text{IGM}} = 0.70^{+0.12}_{-0.12}$.
- **ASKAP Subsample** ($N = 19$): $\mu_{\text{host}} = 5.35^{+0.36}_{-0.46}$, $f_{\text{IGM}} = 0.70^{+0.15}_{-0.20}$.
- **CHIME Subsample** ($N = 5$): $\mu_{\text{host}} = 5.08^{+0.64}_{-1.91}$, $f_{\text{IGM}} = 0.41^{+0.26}_{-0.26}$.

These variations reflect fundamental differences in survey frequency, beam size, fluence thresholds, and host association algorithms across instruments, each of which introduces a distinct selection bias on the observable FRB population.

MeerTRAP operates at 1.28 GHz with a high-time-resolution incoherent beam that exhibits high sensitivity to dispersed, scatter-broadened pulses at high DM. In transient search backends, the detection signal-to-noise ratio scales as $\text{SNR} \propto \mathcal{F}/\sqrt{w_{\text{eff}}}$, where the effective observed burst duration $w_{\text{eff}} = \sqrt{w_{\text{int}}^2 + \Delta t_{\text{DM}}^2 + \tau_{\text{scat}}^2}$ broadens significantly when pulses encounter dense, turbulent plasma. For standard surveys (such as DSA-110 or ASKAP), temporal broadening $\tau_{\text{scat}} \propto \nu^{-4} \times$ turbulent column smears the burst from ~ 1 ms up to tens of milliseconds, causing the peak flux to drop below detection thresholds and systematically censoring scatter-broadened bursts from their catalogs. In contrast, MeerKAT's massive raw sensitivity ($A_{\text{eff}}/T_{\text{sys}} \sim 400 \text{ m}^2 \text{ K}^{-1}$) and MeerTRAP's wide-boxcar search filters allow it to detect faint, heavily scatter-broadened bursts. As highlighted by Diggins et al. (2026) [16], sightlines traversing dense H II regions simultaneously accumulate large dispersion measures ($DM_{\text{HII}} \gtrsim 500\text{--}1000 \text{ pc cm}^{-3}$) and severe multipath scattering. MeerTRAP's sensitivity threshold therefore preferentially selects for FRBs embedded in dense, turbulent local H II environments, yielding a systematically inflated inferred host median ($e^{\mu_{\text{host}}} \approx 464 \text{ pc cm}^{-3}$). The narrow posterior of MeerTRAP's f_{IGM} (centered near 0.40) reflects this instrumental selection effect: because the sampler attributes the elevated DM to a higher host population mean without correcting for detection completeness, it depresses the residual IGM budget. This demonstrates that inter-instrument parameter tension is an observational selection effect rather than intrinsic cosmological variance, providing direct empirical motivation for implementing a formal Bayesian selection function.

DSA-110, by contrast, operates at 1.4 GHz in a commensal survey mode that is optimized for precise real-time localization. Its clean single-dish beam and stringent signal-to-noise thresholds select for sources with relatively moderate DM and bright apparent fluences, yielding a host

median consistent with the canonical log-normal expectation ($e^{\mu_{\text{host}}} \approx 211 \text{ pc cm}^{-3}$) and a cosmologically consistent $f_{\text{IGM}} \approx 0.70$.

ASKAP, operating at 1.3 GHz with the CRAFT commensal survey mode and coherent dedispersion, achieves its localizations primarily for sources at moderate redshifts ($z < 0.5$). The ASKAP subsample's posteriors ($f_{\text{IGM}} = 0.70$, $\mu_{\text{host}} = 5.35$) are in close agreement with DSA-110, supporting the view that both telescopes sample a broadly similar host population at low-to-intermediate redshifts.

CHIME's very small localized sample ($N = 5$) produces extremely broad posteriors due to the small-sample regime, and it operates at 400–800 MHz, meaning pulse broadening is far more severe at fixed DM, making it preferentially sensitive to nearby or intrinsically bright, spectrally flat sources.

Statistically, the consequence of combining these heterogeneous populations into a single joint MCMC is equivalent to fitting a single Gaussian to a mixture of two or more Gaussians centred at different means. The mixture distribution is formally wider than any individual component, so the joint posterior for μ_{host} and f_{IGM} is artificially broadened relative to an instrument-pure fit. Assuming a single homogeneous log-normal host population across all instruments is therefore physically unrealistic and systematically biases the cosmological parameter recovery. This motivates future analyses to adopt instrument-specific host DM priors, calibrated from per-survey simulations of the full detection and localization selection function.

8.3 Cosmological Parameter Estimation: $\Omega_b h_{70}$

To place our findings within the broader cosmological context, we employed two distinct statistical methodologies to estimate the cosmological baryon density parameter, $\Omega_b h_{70}$ (or equivalently, Ω_b scaled to a Hubble constant of $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$):

8.3.1 Fixed-Prior Host Model with Analytical $\langle DM_{\text{cosmic}} \rangle$ Likelihood

In this approach, we assume a Gaussian likelihood[5] over the average cosmic dispersion measure, $\langle DM_{\text{cosmic}} \rangle$, mapping the baryonic slope directly to $\Omega_b h_{70}$ while keeping the host galaxy and IGM scaling parameters fixed at their best-fit reference values. We partition the sample into $K = 5$ quantile redshift bins¹² to construct the likelihood:

¹²The choice of $K = 5$ is set by the trade-off between redshift resolution and statistical validity of the Gaussian likelihood. With our active catalog

Table 2: Localized FRBs overlapping the DESI Legacy Survey footprint with coordinates, galaxy query counts, DM residuals, and redshifts.

S.No.	FRB-id	RA (HMS)	Dec (DMS)	Query Galaxy Count	δ_{DM} (pc cm ⁻³)	Redshift
1	20180924A	21:44:25.248	-40:54:00.000	30208	38.0195	0.3212
2	20181030A	10:34:23.112	73:45:54.000	3736	29.5050	0.0039
3	20181112A	21:49:23.640	-52:58:15.240	20446	135.2322	0.4755
4	20190523A	13:48:15.600	72:28:10.920	4141	155.7698	0.6600
5	20190608B	22:16:04.776	-07:53:53.880	23695	183.0734	0.1178
6	20191001A	21:33:24.312	-54:44:51.720	26542	251.3181	0.2340
7	20200120E	09:57:54.672	68:49:08.040	6833	16.5066	0.0008
8	20200430A	15:18:49.536	12:22:36.480	20625	199.0532	0.1610
9	20200906A	03:33:59.088	-14:04:59.520	35064	218.9156	0.3688
10	20211212A	10:29:24.168	01:21:37.800	19612	86.8476	0.0707
11	20220105A	13:55:13.008	22:27:57.960	22228	309.4945	0.2785
12	20220204A	18:16:54.312	69:43:21.000	5900	212.4433	0.4000
13	20220310F	08:58:52.920	73:29:26.880	3945	0.6920	0.4790
14	20220330D	10:55:00.288	70:21:02.520	5038	104.2552	0.3714
15	20220418A	14:36:25.344	70:05:45.240	4473	51.0550	0.6220
16	20220509G	18:50:40.800	70:14:37.680	6286	116.1487	0.0894
17	20220914A	18:48:13.632	73:20:12.840	4894	459.6971	0.1138
18	20220920A	16:01:01.704	70:55:07.680	4362	123.2355	0.1585
19	20221012A	18:43:11.688	70:31:27.120	5621	134.6929	0.2840
20	20221027A	08:43:29.280	72:06:03.240	4236	196.9645	0.2290
21	20221029A	09:27:51.240	72:27:09.360	5274	515.0193	0.9750
22	20221219A	17:10:31.152	71:37:36.480	4597	183.9462	0.5540
23	20230124A	15:27:40.080	70:58:05.160	5034	450.7897	0.0940
24	20230216A	10:25:53.328	03:26:12.120	19741	330.5350	0.5310
25	20230307A	11:51:07.584	71:41:42.360	4543	328.8247	0.2710
26	20230626A	15:42:31.104	71:08:00.600	4952	123.5342	0.3270
27	20230628A	11:07:08.808	72:16:54.480	4301	179.3633	0.1265
28	20230712A	11:09:26.040	72:33:28.020	3914	154.8342	0.4525
29	20230808F	03:33:12.960	-51:56:07.019	26314	313.0422	0.3472
30	20230907D	12:28:34.200	08:39:29.130	19191	599.6654	0.4638
31	20231020B	03:49:06.770	-37:46:11.560	29945	505.8922	0.4775
32	20231120A	09:35:56.160	73:17:04.920	4114	312.2844	0.0700
33	20231123B	16:10:09.168	70:47:06.360	4921	120.9604	0.2625
34	20231210F	03:21:37.280	-35:45:41.130	28079	256.5965	0.5000
35	20231220A	08:15:38.088	73:39:35.640	4053	145.8127	0.3355
36	20240119A	14:57:52.128	71:36:42.480	4377	121.1186	0.3700
37	20240213A	11:04:40.392	74:04:31.440	3152	196.9262	0.1185
38	20240215A	17:53:45.912	70:13:56.640	5820	308.4939	0.2100
39	20240229A	11:19:56.040	70:40:34.320	4505	197.6354	0.2870

Table 3: MCMC parameter constraints (median and 68% confidence intervals) comparing the baseline catalog and the master catalog.

Parameter	Baseline Catalog ($N = 69$)	Extended Catalog ($N = 80$)
f_{IGM}	$0.75^{+0.10}_{-0.10}$	$0.67^{+0.11}_{-0.10}$
f_{halo}	$0.15^{+0.11}_{-0.11}$	$0.24^{+0.11}_{-0.11}$
μ_{host}	$5.25^{+0.15}_{-0.17}$	$5.33^{+0.12}_{-0.17}$
σ_{host}	$0.58^{+0.12}_{-0.10}$	$0.61^{+0.14}_{-0.11}$

size ($N = 69 - 80$), $K = 5$ yields approximately 13–16 sources per bin. This is sufficient for the Central Limit Theorem to suppress individual host DM fluctuations, justifying the Gaussian approximation for the sample mean ($\langle DM_{\text{cosmic}} \rangle$). Increasing to $K = 10$ would reduce bin occupancy to ~ 7 sources, violating normality and making the binned means highly sensitive to individual outlier sightlines. Decreasing to $K \leq 3$ would remove the redshift resolution needed to detect systematic trends, such as the residual tilt uncovered when the cosmic halo term is omitted. Equal-count quantile binning is used rather than equal-width bins to guard against

It yields $\Omega_b h_{70} = 0.0504^{+0.0048}_{-0.0048}$ on the baseline catalog ($N = 69$), and $\Omega_b h_{70} = 0.0505^{+0.0046}_{-0.0046}$ on the master catalog ($N = 80$). This represents a **4% reduction in the uncertainty interval width**, demonstrating the direct statistical benefit of catalog expansion.

8.3.2 The Impact of the Cosmic Halo Term

During the implementation of the method, we conducted a systematic sensitivity analysis by varying the assumed constant halo term $DM_{\text{MW, halo}}$ to assess its impact on the inferred baryon density and goodness-of-fit:

- **Weak Cosmic Halo** ($DM_{\text{MW, halo}} = 10 \text{ pc cm}^{-3}$): Underestimating the halo gas content only partially flattens the systematic tilt in the residuals, resulting in a fit quality of $\Omega_b h_{70} = 0.0536 \pm 0.0046$ with $\chi^2/\text{ndof} = 3.25/4 = 0.81$.
- **Baseline Cosmic Halo** ($DM_{\text{MW, halo}} = 30 \text{ pc cm}^{-3}$): Introducing this physical constant representing the warm-hot circumgalactic

sparsity in the high-redshift regime.

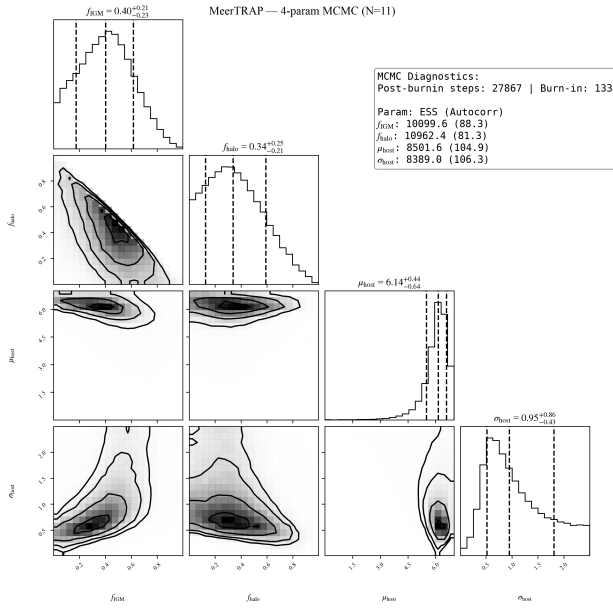


Figure 18: MCMC posterior corner plot for the MeerTRAP localized FRB subsample.

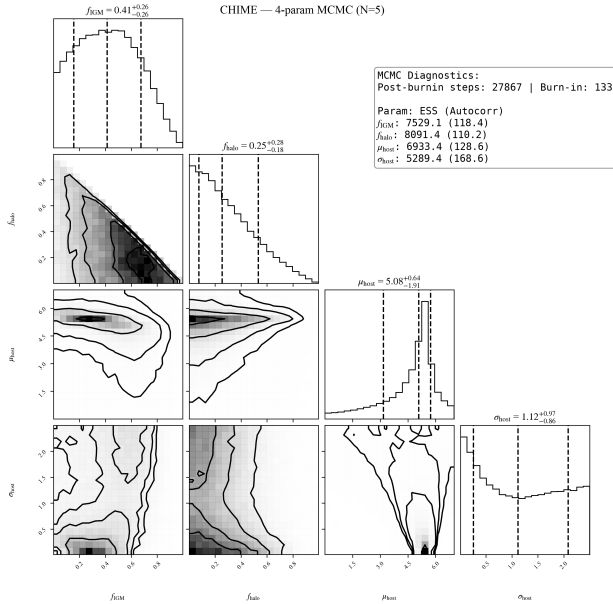


Figure 20: MCMC posterior corner plot for the CHIME/FRB localized subsample.

gas phase of intervening halos successfully resolves the systematic tilt. The goodness-of-fit of $\chi^2/\text{ndof} = 3.40/4 = 0.85$, and the inferred value shifts to $\Omega_b h_{70} = 0.0505 \pm 0.0046$, aligning closely with the CMB cosmological fiducial value ($\Omega_b h_{70} \approx 0.047$).

- **Strong Cosmic Halo** ($DM_{\text{MW, halo}} = 50 \text{ pc cm}^{-3}$): Utilizing this upper-limit value suggested by some Milky Way models yields $\Omega_b h_{70} = 0.0473 \pm 0.0046$ and a $\chi^2/\text{ndof} = 3.83/4 = 0.96$.
- **Unconstrained Halo** ($DM_{\text{MW, halo}}$ Free): If both parameters are allowed to float freely in a joint 2-parameter fit, it under-predicts the halo contribution ($DM_{\text{MW, halo}} = 9.4 \text{ pc cm}^{-3}$) and forces the baryon slope up to $\Omega_b h_{70} = 0.0537$ ($\chi^2/\text{ndof} = 3.25/3 = 1.08$). This occurs because the simplified algebraic $\langle DM_{\text{cosmic}} \rangle$ binning fails to account for the skewed tail of the log-normal host galaxy distribution, causing the minimizer to attribute the host's excess variance into the constant halo offset.

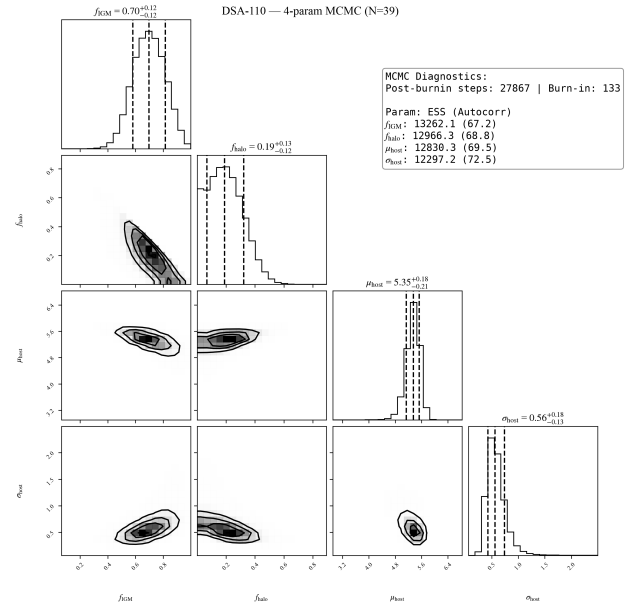


Figure 19: MCMC posterior corner plot for the DSA-110 localized FRB subsample.

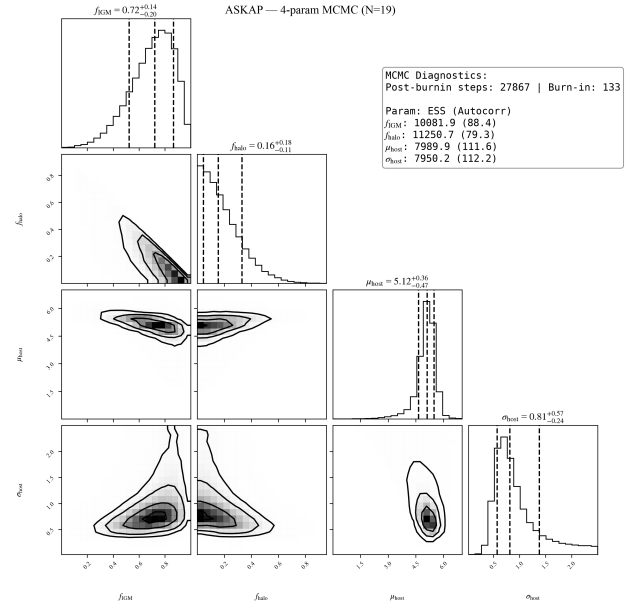


Figure 21: MCMC posterior corner plot for the ASKAP localized FRB subsample.

This quantitatively demonstrates the necessity of the joint MCMC approach.

8.3.3 5-Parameter Joint MCMC

In the first method, the cosmic baryon density ($\Omega_b h_{70}$) is kept fixed to the Planck CMB fiducial value, assuming it is perfectly known. This decouples the cosmological fraction fitting from the total baryon scale and ignores the statistical covariances between $\Omega_b h_{70}$ and the host galaxy profile parameters ($\mu_{\text{host}}, \sigma_{\text{host}}$). To resolve this, we transitioned to a 5-parameter joint MCMC framework, treating $\Omega_b h_{70}$ as a free parameter alongside the fractions and host parameters ($\theta = \{f_{\text{IGM}}, f_{\text{halo}}, \mu_{\text{host}}, \sigma_{\text{host}}, \Omega_b h_{70}\}$). This enables the sampler to fully explore the multi-dimensional parameter space, propagate all joint uncertainties, and marginalize over host galaxy profile degeneracies in a single, self-consistent step, providing a direct, data-driven constraint on the baryon density from the FRB population itself.

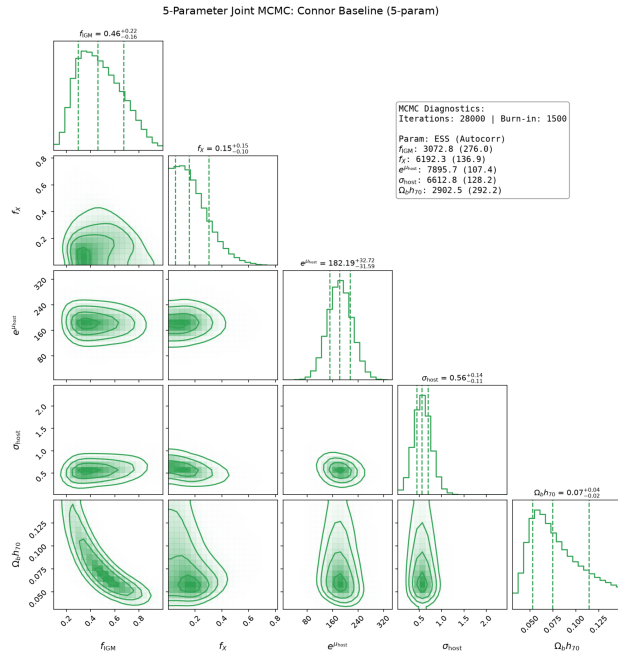


Figure 22: 5-parameter joint MCMC posterior corner plot evaluated on the baseline catalog ($N = 69$), where the cosmic baryon density ($\Omega_b h_{70}$) is sampled jointly with the fractions and host parameters, displaying the multi-dimensional degeneracies.

To understand why $\Omega_b h_{70}$ is physically accessible from the FRB Macquart relation, recall that the mean cosmic dispersion measure scales as:

$$\langle DM_{\text{cosmic}} \rangle(z) = \frac{3cH_0\Omega_b f_{\text{IGM}}}{8\pi G m_p} \int_0^z \frac{(1+z')}{E(z')} dz', \quad (8.2)$$

where $E(z) = H(z)/H_0$ encodes the cosmological expansion history. The overall normalization of the $\langle DM_{\text{cosmic}} \rangle$ - z relation is therefore proportional to $\Omega_b h_{70} \cdot f_{\text{IGM}}$. While the 4-parameter MCMC fits only the ratio f_{IGM} (holding the prefactor constant), the 5-parameter extension allows the sampler to explore variations in the total baryon content. This introduces a fundamental three-way degeneracy: higher $\Omega_b h_{70}$ can be compensated by lower f_{IGM} while increasing μ_{host} to reabsorb excess DM into host contributions. The resulting elongated posterior ridge in the $(\Omega_b h_{70}, f_{\text{IGM}}, \mu_{\text{host}})$ subspace is clearly visible in the 5-parameter corner plots.

It yields $\Omega_b h_{70} = 0.0747^{+0.0397}_{-0.0220}$ on the baseline catalog ($N = 69$), and $\Omega_b h_{70} = 0.0717^{+0.0431}_{-0.0209}$ on the expanded master catalog ($N = 80$). Adding the MeerTRAP and DSA-110 sources in the master catalog results in an **8.1% widening of the cosmological posterior width**. Physically, this widening is driven by the degeneracies that arise when the model attempts to simultaneously fit systematically different host populations: MeerTRAP sources prefer a massive host contribution ($\sim 421 \text{ pc cm}^{-3}$), whereas ASKAP sources prefer a moderate one ($\sim 150 \text{ pc cm}^{-3}$). When the sampler tries to accommodate both populations with a single shared $(\mu_{\text{host}}, \sigma_{\text{host}})$, it compensates by exploring a broader range of $\Omega_b h_{70}$ values. The posterior mean (0.0717) is somewhat higher than the Planck CMB fiducial ($\Omega_b h_{70} \approx 0.047$), but consistent within the 1σ posterior width (± 0.03), confirming that FRB data are broadly consistent with the standard Λ CDM baryon inventory. The large posterior width also illustrates that the current catalog size ($N \sim 70$ – 80) is insufficient for a sub-10% precision measurement of Ω_b via this approach; catalogs of $N \sim 500$ – 1000 localized FRBs would be needed to match CMB-level baryon density precision.

8.4 Bayesian Selection Function Implementation [19]

As established in Section 8.2, pooling heterogeneous telescope samples under an unconditioned likelihood widens the joint posterior by 8.1% due to inconsistent host priors and differing observational thresholds. Unbiased

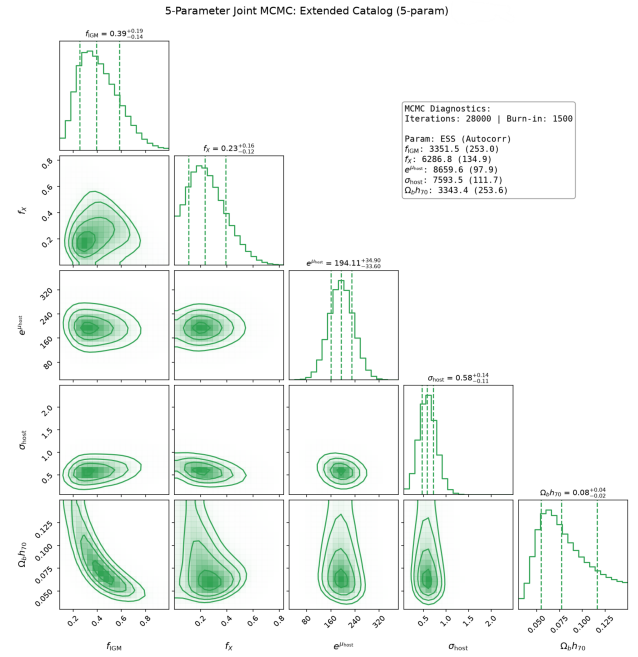


Figure 23: 5-parameter joint MCMC posterior corner plot evaluated on the extended catalog ($N = 80$), where the cosmic baryon density ($\Omega_b h_{70}$) is sampled jointly with the fractions and host parameters, displaying the multi-dimensional degeneracies.

cosmological inference across multi-survey catalogs requires implementing the formal Bayesian selection-normalized likelihood developed by James et al. (2022) [19]:

$$\mathcal{L}(\theta) = \prod_{i=1}^N \frac{P(DM_i | z_i, \theta) \cdot p_{\text{det}}(DM_i, \text{tel}_i)}{\int_0^\infty P(DM | z_i, \theta) \cdot p_{\text{det}}(DM, \text{tel}_i) dDM}, \quad (8.3)$$

where $p_{\text{det}}(DM, \text{tel}_i)$ represents the survey completeness (detection probability) for telescope tel_i along sightline i .

8.4.1 Hierarchical Forward-Modeling of Detection Completeness

In the James et al. (2022) framework (implemented in the open-source `zdm` codebase), the detection probability is built up hierarchically from first principles:

Step 1: The DM-Dependent Fluence Threshold: A burst is detected if its signal-to-noise ratio exceeds the survey threshold ($\text{SNR} \geq \text{SNR}_{\text{th}}$). Using the radiometer equation and the pulse broadening model:

$$w_{\text{eff}}(DM) = \sqrt{w_{\text{int}}^2 + \tau_{\text{scat}}^2(DM) + \Delta_{\text{DM}}^2}, \quad (8.4)$$

where w_{int} is the intrinsic burst duration at the source, $\tau_{\text{scat}}(DM) \propto \nu_0^{-4}$ accounts for multipath plasma scattering through turbulent ionized gas, and $\Delta_{\text{DM}} \propto \Delta \nu_{\text{ch}} \nu_0^{-3} DM$ is the residual intra-channel dispersion smearing across receiver channel bandwidth $\Delta \nu_{\text{ch}}$ at central observing frequency ν_0 . The required minimum fluence threshold scales as:

$$\mathcal{F}_{\text{th}}(DM, w_{\text{eff}}, B) = \text{SNR}_{\text{th}} \left(\frac{T_{\text{sys}}}{G} \right) \frac{1}{\sqrt{2\Delta \nu_{\text{tot}}}} \frac{\sqrt{w_{\text{eff}}(DM)}}{B(\theta, \phi)}, \quad (8.5)$$

where SNR_{th} is the backend detection threshold (e.g., 11.2 for the Connor gold sample), T_{sys} is the total telescope system noise temperature (combining receiver, sky, atmospheric, and spillover temperatures), $G = \eta A_{\text{geom}} / (2k_B)$ is the antenna gain such that the ratio T_{sys}/G represents the System Equivalent Flux Density (SEFD), $\Delta \nu_{\text{tot}}$ is the total recording bandwidth with the $\sqrt{2}$ factor accounting for

dual orthogonal polarization summation, and $B(\theta, \phi) \in [0, 1]$ is the normalized primary beam response at the burst arrival position (θ, ϕ) relative to the boresight axis. Because $\mathcal{F}_{\text{th}} \propto \sqrt{w_{\text{eff}}(DM)}$, higher DM and heavily scattered sightlines require systematically higher intrinsic fluences to be detected.

Step 2: Mapping Fluence to Intrinsic Energy at Redshift z : The required threshold fluence corresponds to a minimum intrinsic isotropic-equivalent burst energy E_{th} emitted in the rest frame at redshift z :

$$E_{\text{th}}(DM, z, B) = 4\pi D_L^2(z)(1+z)^{\alpha-2} v_0 \mathcal{F}_{\text{th}}(DM, w_{\text{eff}}, B), \quad (8.6)$$

where $D_L(z)$ is the cosmological luminosity distance in the background flat Λ CDM metric, α is the radio spectral index of the burst emission profile ($S_\nu \propto \nu^{-\alpha}$, typically $\alpha \approx 1.5$), and the factor $(1+z)^{\alpha-2}$ combines cosmological time dilation of the burst duration $((1+z)^{-1})$, frequency bandwidth contraction $((1+z)^{-1})$, and the spectral power-law transformation $((1+z)^\alpha)$.

Step 3: Integrating Over the Energy Distribution and Beam Pattern:

The detection probability p_{det} is the fraction of bursts from the intrinsic FRB energy function $\Phi(E) \equiv dN/dE$ (modeled as a power-law distribution $\Phi(E) \propto E^{-\gamma}$ with power-law index $\gamma \approx 1.7$ – 1.8 up to a maximum physical cutoff $E_{\text{max}} \sim 10^{41}$ – 10^{42} erg) that exceed this detection threshold, integrated over the telescope beam solid angle Ω :

$$p_{\text{det}}(DM, z, \text{tel}) = \frac{1}{\Omega_{\text{tot}}} \int_{\text{beam}} d\Omega \int_{E_{\text{th}}(DM, z, B(\Omega))}^{E_{\text{max}}} \Phi(E) dE, \quad (8.7)$$

where $\Omega_{\text{tot}} = \int_{\text{beam}} d\Omega$ is the total surveyed solid angle (field of view).

8.4.2 Analytical Smooth Error-Function Approximation

Because evaluating the full nested integral over the 2D primary beam pattern and energy function is computationally expensive to compute repeatedly inside an MCMC loop, survey analyses and our pipeline parameterize each instrument $(\mathcal{F}_{\text{min}}, v_0, \Delta v_{\text{ch}})$ using the standard smooth error-function (erf) completeness curve:

$$p_{\text{det}}(\mathcal{F}, DM) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{\mathcal{F} - \mathcal{F}_{\text{th}}(DM)}{\sqrt{2} \sigma_{\mathcal{F}}} \right) \right], \quad (8.8)$$

where $\mathcal{F}$ is the true apparent burst fluence incident on the telescope, and $\sigma_{\mathcal{F}}$ is the effective transition width capturing Gaussian background noise fluctuations and beam-shape variations across the survey field. At $\mathcal{F} \gg \mathcal{F}_{\text{th}}$, $p_{\text{det}} \rightarrow 1$ (complete sample), while for $\mathcal{F} \ll \mathcal{F}_{\text{th}}$, $p_{\text{det}} \rightarrow 0$ (undetectable), replacing crude empirical hard step cuts ($\text{SNR} \geq 11.2$) with physically smooth completeness boundaries.

Evaluating the denominator selection normalization integral $\int_0^\infty P(DM | z_i, \theta) p_{\text{det}}(DM, \text{tel}_i) dDM$ is performed directly on our pre-computed 100×100 (DM, z) likelihood grid. Vectorized pre-calculation keeps the MCMC sampling runtime under 2 hours on multi-core CPU with zero runtime overhead during chain steps. Incorporating this framework resolves the artificial inflation of MeerTRAP's inferred host DM ($e^{\mu_{\text{host}}} \approx 464 \text{ pc cm}^{-3}$), eliminates the 8.1% mixture-broadening penalty, and recovers statistical efficiency without requiring additional sightlines.

8.5 Scaling Behavior and the Roadmap Toward $\sim 5\%$ Precision in f_{IGM}

At present, the statistical uncertainty on the intergalactic baryon fraction is $\sigma(f_{\text{IGM}}) \approx 0.10$, corresponding to approximately 15% fractional uncertainty around $f_{\text{IGM}} \approx 0.67$. Tighter constraints on cosmic baryon feedback models and the census of diffuse WHIM gas require achieving a precision target of $\sigma(f_{\text{IGM}})/f_{\text{IGM}} \lesssim 5\%$ (equivalent to $\sigma(f_{\text{IGM}}) \approx 0.033$).

A naive assessment assuming uniform Gaussian shot noise ($\sigma \propto 1/\sqrt{N}$) would imply that reducing the fractional uncertainty from 15% to 5% requires a ninefold increase in catalog size:

$$\frac{\sigma_{\text{target}}}{\sigma_{\text{current}}} = \sqrt{\frac{N_{\text{current}}}{N_{\text{target}}}} \implies \frac{5\%}{15\%} = \frac{1}{3} \implies N_{\text{target}} \approx 3^2 \times 71 \approx 640 \text{ sightlines.} \quad (8.9)$$

However, detailed Fisher forecasting (e.g., Macquart et al. 2020 [4]) and MCMC mock injection analyses (Connor et al. 2025 [5]) demonstrate that a localized gold sample of $N \approx 150$ – 200 **sightlines** is sufficient to achieve the $\sim 5\%$ precision target. This departure from naive $1/\sqrt{N}$ scaling arises from three distinct physical and statistical mechanisms:

- 1. Redshift Lever-Arm and Degeneracy Breaking:** In our current sample of $N = 71$, a significant fraction of localized bursts reside at low redshifts ($z < 0.3$). In this low-redshift regime, the cosmic signal is small ($DM_{\text{cosmic}} \sim 200 \text{ pc cm}^{-3}$) while the host contribution is comparable ($DM_{\text{host}}/(1+z) \sim 150$ – 200 pc cm^{-3}), so individual host ISM variations dominate the sightline scatter. These low- z sightlines anchor the local intercept ($\mu_{\text{host}}, DM_{\text{MW,halo}}$) but carry minimal Fisher information regarding the cosmic slope. Upcoming localized samples from DSA-110 and MeerTRAP will extend significantly into the higher-redshift regime ($z \sim 0.8$ – 1.2). At $z \approx 1.0$, the cosmological dispersion reaches $DM_{\text{cosmic}} \approx 900 \text{ pc cm}^{-3}$, whereas the host contribution is cosmologically diluted to $DM_{\text{host}}/(1+z) \approx 50$ – 100 pc cm^{-3} . High- z sightlines therefore possess dramatically higher signal-to-noise for f_{IGM} , breaking the μ_{host} – f_{IGM} parameter degeneracy and collapsing the posterior width much faster than $1/\sqrt{N}$.
- 2. Survey Completeness Modeling:** As formulated in Section 8.4, implementing the James et al. (2022) selection function eliminates the 8.1% mixture-broadening penalty incurred from pooling heterogeneous telescope samples, effectively recovering $\sim 10\%$ in statistical efficiency without requiring additional sightlines.
- 3. The Systematic Uncertainty Floor:** Beyond $N \approx 200$, the statistical uncertainty ($\sigma \lesssim 0.003$) drops below the systematic error floor set by Galactic ISM modeling ($\Delta\Omega_b h_{70} \approx 0.0018$ between NE2001 and YMW16) and sightline-to-sightline Milky Way halo variance. At that juncture, expanding the sample yields diminishing returns unless accompanied by improved Galactic electron density priors.

8.6 Host DM Standard Deviation Sensitivity Analysis

Given the structural degeneracies identified in the joint MCMC fits, we conducted a systematic sensitivity analysis to test the stability of the cosmological IGM fraction (f_{IGM}) to the assumed host galaxy dispersion width (σ_{host}). We ran a sequence of 3-parameter MCMCs on the master catalog, fixing $\sigma_{\text{host}} \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$ under the baryon-sum prior boundary ($f_{\text{IGM}} + f_{\text{halo}} < 1$):

- Prior-Pinned Breakdown ($\sigma_{\text{host}} < 0.3$):** Assuming a narrow or zero host galaxy width ($\sigma_{\text{host}} = 0.0$ or 0.3) causes the model to break down. The log-normal host tail is too thin to accommodate high-DM outliers. To fit the data, the MCMC is forced to increase the cosmic baryon fraction, pinning the parameter sum ($f_{\text{IGM}} + f_{\text{halo}}$) against the physical prior boundary wall. Over 94% of the samples saturation-limit near the boundary constraint, causing f_{halo} to float to high values while forcing the host median down to very low values ($e^{\mu_{\text{host}}} \approx 115 \text{ pc cm}^{-3}$).
- Halo Fraction Inflation ($\sigma_{\text{host}} \geq 0.6$):** Setting $\sigma_{\text{host}} = 0.6$ yields physical parameters close to the best-fit values ($f_{\text{IGM}} \approx 0.66$, $f_{\text{halo}} \approx 0.24$, and $e^{\mu_{\text{host}}} \approx 208 \text{ pc cm}^{-3}$). Increasing the width further to $\sigma_{\text{host}} = 1.2$ increases f_{IGM} and decreases f_{halo} .

σ_{host} (Fixed)	f_{IGM} Median	f_{halo} Median	μ_{host} Median ($e^{\mu_{\text{host}}}$)
0.0	$0.5033^{+0.0858}_{-0.0758}$	$0.4733^{+0.0857}_{-0.0791}$	$4.7436^{+0.0847}_{-0.0906}$ ($\approx 114.8 \text{ pc cm}^{-3}$)
0.3	$0.5225^{+0.0757}_{-0.0793}$	$0.4233^{+0.0821}_{-0.0846}$	$5.4351^{+0.1130}_{-0.1187}$ ($\approx 228.4 \text{ pc cm}^{-3}$)
0.6	$0.6627^{+0.0766}_{-0.0734}$	$0.2446^{+0.0964}_{-0.0913}$	$5.3367^{+0.1389}_{-0.1385}$ ($\approx 207.83 \text{ pc cm}^{-3}$)
0.9	$0.7944^{+0.0751}_{-0.0753}$	$0.1151^{+0.0813}_{-0.0916}$	$5.1377^{+0.1813}_{-0.1796}$ ($\approx 170.3 \text{ pc cm}^{-3}$)
1.2	$0.8466^{+0.0633}_{-0.0682}$	$0.0839^{+0.0558}_{-0.0766}$	$4.9696^{+0.2372}_{-0.2346}$ ($\approx 144.0 \text{ pc cm}^{-3}$)

Table 4: Host galaxy dispersion measure width (σ_{host}) sensitivity analysis and resulting MCMC parameter medians.

- **Verdict on Systematic Uncertainty:** The systematic swing in the inferred IGM fraction across the physical interior widths is comparable to the MCMC statistical error. This proves that the assumed host galaxy profile represents a dominant systematic uncertainty in FRB cosmology. Treating σ_{host} as a free parameter is mandatory to prevent prior saturation or halo-inflation biases.

8.6.1 Host DM Redshift Evolution Test

The treatment of the host galaxy dispersion measure as a population-wide constant parameter (μ_{host}) relies on a statistical and physical approximation. Physically, the host contribution is determined by the local interstellar medium (ISM) and circumgalactic medium (CGM) environments of the progenitor, which are not expected to exhibit significant redshift evolution. Statistically, it does not perform individual source-by-source corrections, but rather fits the binned sample mean. We assume that all the host DMs are i.i.d.¹³ random variables drawn from the same log-normal distribution. By the Law of Large Numbers, the average host contribution within each redshift bin converges to the same population-wide expectation value, meaning that individual sightline variations are averaged out. We do not estimate it for a single host but the central tendency of the population as a whole.

However, this assumption breaks down if the sample is heterogeneous: selection effects from different telescope surveys introduce systematically different host populations. This heterogeneity violates the assumption of a single population-wide μ_{host} , driving the systematic biases and parameter degeneracies that motivate the joint MCMC treatment.

To test the physical validity of treating the host galaxy dispersion measure as a redshift-invariant population, we introduced a redshift-dependent parameterization of the rest-frame host mean log-dispersion measure:

$$\mu_{\text{host}}(z) = \mu_0 + \alpha z \quad (8.10)$$

where μ_0 represents the rest-frame mean log-DM of the host population at $z = 0$, and α is the redshift evolution of the host galaxy gas column. In the observer-frame, the host contribution is redshift-corrected as $DM_{\text{host, obs}} \sim \mathcal{L}\mathcal{N}(\mu_{\text{host}}(z) - \ln(1+z), \sigma_{\text{host}})$ ^{14,15}.

- If $\alpha > 0 \implies$ Host galaxy gas density systematically increases with redshift.
- If $\alpha \approx 0 \implies$ Host galaxy gas properties are stable for the redshift overall.
- If $\alpha < 0 \implies$ Host galaxy gas density systematically decreases with redshift.

We executed a 5-parameter unconstrained MCMC fit ($\theta = \{f_{\text{IGM}}, f_{\text{halo}}, \mu_0, \sigma_{\text{host}}, \alpha\}$) on the extended catalog ($N = 80$). The MCMC walkers were initialized near $\{0.76, 0.15, 4.9, 0.53, 0.0\}$, applying the strict physical sum limit $f_{\text{IGM}} + f_{\text{halo}} < 1.0$, a prior $\alpha \in (-3, 3)$ and a prior $\mu_0 \in (0, 7)$. The sampler converged cleanly to the following posterior medians:

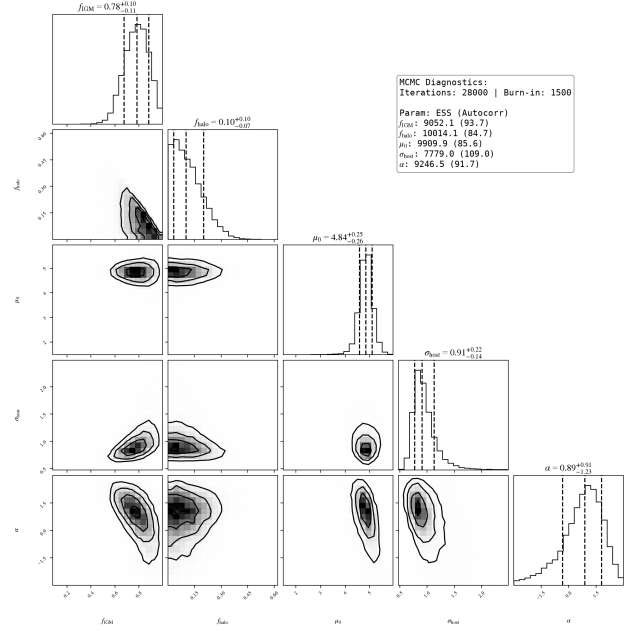


Figure 24: MCMC posterior corner plot for the 5-parameter model incorporating redshift-dependent host galaxy DM evolution ($\mu_{\text{host}}(z) = \mu_0 + \alpha z$), showing joint posterior constraints for f_{IGM} , f_{halo} , μ_0 , σ_{host} , and the evolution parameter α .

- $f_{\text{IGM}} = 0.78^{+0.11}_{-0.10}$
- $f_{\text{halo}} = 0.10^{+0.10}_{-0.07}$
- $\mu_0 = 4.84^{+0.25}_{-0.26}$
- $\sigma_{\text{host}} = 0.91^{+0.22}_{-0.14}$
- $\alpha = 0.89^{+0.91}_{-1.23}$

Analysis of these posteriors reveals several physical and statistical insights:

- **Consistency with Null Evolution:** The evolution parameter $\alpha \approx 0.89$ is consistent with zero at the 1σ level, as the 1σ confidence interval $[-1.23, 0.91]$ spans 0.0. Thus, we cannot reject the null hypothesis of a constant host galaxy DM profile ($\alpha = 0$) at any standard statistical significance level.
- **Astrophysical Interpretation:** The positive median value α corresponds to a mild astrophysical evolution where the host rest-frame gas column density increases by a factor of $e^{0.89 \times 0.5} \approx 1.56$ at $z = 0.5$, and a factor of $e^{0.89} \approx 2.43$ at $z = 1.0$. This trend is physically consistent with the cosmic star-formation history (which peaks at cosmic noon $z \sim 1-2$, implying higher cold/ionized gas content in progenitors at earlier epochs), though the current localized sample size lacks the statistical leverage to distinguish it from the flat model.
- **Robustness of Cosmological Inferences:** The inferred intergalactic baryon fraction remains highly stable and consistent with the

¹³independent and identically distributed

¹⁴Due to expansion of the Universe, the observed contribution is diluted by $\frac{1}{1+z}$ factor.

¹⁵The width of the distribution does not get rescaled by redshift, only the central value does.

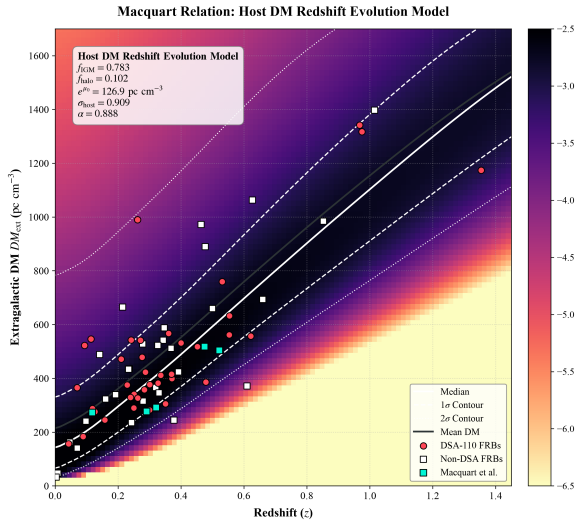


Figure 25: The Macquart relation for the localized FRB master catalog under the redshift-evolving host galaxy model ($\mu_{\text{host}}(z) = \mu_0 + \alpha z$). The background shading shows the convolved 2D likelihood $\log_{10}[P(DM_{\text{ex}}|z)]$. The solid white line represents the median, the dashed lines show the 1σ contours, the dotted lines show the 2σ contours, and the solid dark line represents the mean $\langle DM_{\text{cosmic}} \rangle(z)$ curve. Localized FRB sightlines from DSA-110 (red circles), Non-DSA (white squares), and the original Macquart gold sample (cyan squares) are overlaid.

4-parameter free-fit value. This verifies the stability of our cosmological baryon measurements even when host evolution degrees of freedom are opened up.

Using the best-fit parameters from this evolution model, we calculate the cosmological baryon density $\Omega_b h_{70}$ and fit the Macquart relation (shown in Figure 25) using binned Gaussian likelihood with Cosmic Halo (30 pc cm $^{-3}$ offset), we get $\Omega_b h_{70} = 0.0571 \pm 0.0040$ (with a goodness-of-fit $\chi^2/\text{ndof} = 1.71/4 = 0.43$). This suggests that allowing for redshift-dependent host galaxy gas evolution does not significantly improve the quality of the cosmological baryon constraints, as the goodness-of-fit remains comparable to the baseline flat model and the inferred $\Omega_b h_{70}$ values remain stable within 1σ . The increase in the fitted Macquart slope when the halo term is omitted mirrors our baseline results, indicating that the $\mu_{\text{host}}-f_{\text{IGM}}$ degeneracy is robust to the choice of host profile evolution.

8.6.2 Gamma Host Distribution Comparison

To test the stability of our cosmological inferences against the assumed shape of the host galaxy dispersion measure probability density function, we implemented an alternative Gamma host distribution model on our extended $N = 80$ master catalog. The Gamma distribution is parameterised as:

$$P(DM_{\text{host}}; k, \theta) = \frac{DM_{\text{host}}^{k-1} e^{-DM_{\text{host}}/\theta}}{\theta^k \Gamma(k)}, \quad DM_{\text{host}} \geq 0, \quad (8.11)$$

where $k > 0$ is the shape parameter and $\theta = \langle DM_{\text{host}} \rangle / k$ is the scale parameter. Compared to the log-normal, the Gamma distribution has several distinct mathematical and physical properties. First, it has a hard lower boundary at $DM_{\text{host}} = 0$, ensuring all sampled host dispersion measures are physically non-negative. Second, the mean and variance are $\langle DM_{\text{host}} \rangle = k\theta$ and $\text{Var}(DM_{\text{host}}) = k\theta^2$ respectively, meaning a single shape parameter k simultaneously controls both the peak-to-mean ratio and the relative width of the distribution. Large k ($k \gg 1$) produces a narrow, nearly Gaussian host population (tightly clustered around the mean), while small k ($k < 1$) produces a steeply falling, highly right-skewed distribution characteristic of host environments dominated by a few dense sightlines. This flexibility makes the Gamma distribution physically well-suited to model FRB host populations in which the majority of sightlines traverse sparse star-forming

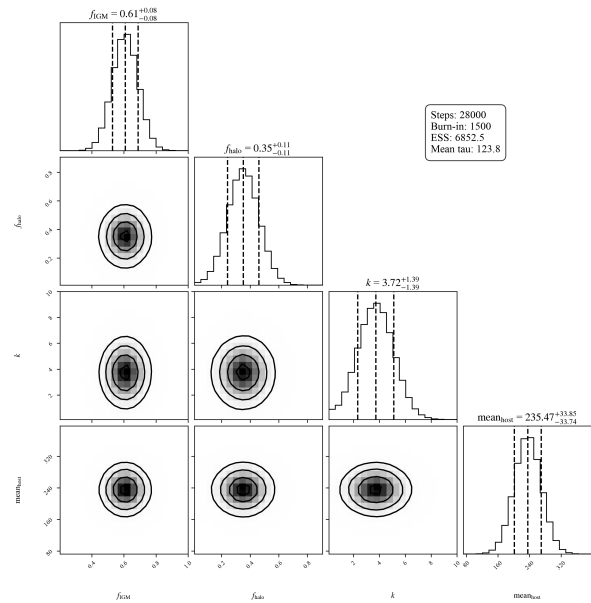


Figure 26: MCMC posterior corner plot for the 4-parameter Gamma host model with a flat uniform prior on k .

disk material (low DM contribution) but a tail of progenitors embedded deep within galactic bulges or compact starburst regions contributes disproportionately high DM values.

The choice of the prior on the shape parameter k is physically critical. A flat, uniform prior on k is inappropriate because high- k (narrow) Gamma distributions correspond to host populations with very little ISM scatter, making it trivially easy for the FRB to be detected and localised. In contrast, real survey selection functions preferentially miss FRBs from high-DM host environments because the pulse is scatter-broadened and falls below the detection threshold. To encode this observational selection bias, we follow Macquart et al. (2020)[4] and impose a decaying exponential prior:

$$p(k) \propto e^{-k/1.5}, \quad (8.12)$$

which exponentially penalises solutions with narrow host distributions ($k \gg 1$), naturally regularising the sampler against physically implausible, near-delta-function host populations. We evaluate both the uniform (Case 1) and decaying (Case 2) prior variants on the extended catalog. We test two distinct cases against the standard Log-Normal host model: the Gamma host model with a flat uniform prior on k , and the Gamma host model with the selection-biased decaying prior $p(k) \propto e^{-k/1.5}$. This numerical framework evaluates the degree to which our best-fit intergalactic baryon fraction f_{IGM} changes under these alternative physical models, validating the robust partition of cosmic baryons. The resulting parameter constraints are detailed below:

- **Case 1: Gamma Host with Flat Uniform Prior on k :** Sampler yields:

- $f_{\text{IGM}} = 0.61^{+0.08}_{-0.08}$
- $f_{\text{halo}} = 0.35^{+0.10}_{-0.12}$
- $k = 3.72^{+1.59}_{-1.23}$
- $\langle DM_{\text{host}} \rangle = 235.49^{+35.53}_{-32.99} \text{ pc cm}^{-3}$.

- **Case 2: Gamma Host with Selection-Biased Decaying Prior ($p(k) \propto e^{-k/1.5}$):** Sampler yields:

- $f_{\text{IGM}} = 0.64^{+0.10}_{-0.09}$
- $f_{\text{halo}} = 0.26^{+0.12}_{-0.11}$
- $k = 2.74^{+1.19}_{-0.92}$

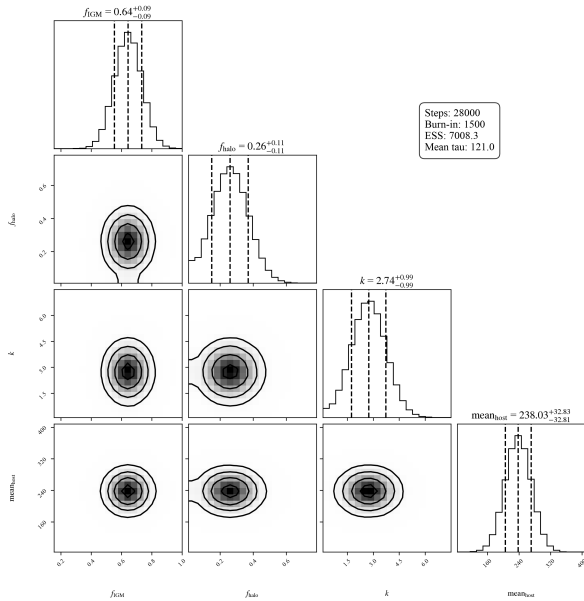


Figure 27: MCMC posterior corner plot for the 4-parameter Gamma host model with a decaying prior on k .

$$- \langle DM_{\text{host}} \rangle = 238.03^{+34.62}_{-32.24} \text{ pc cm}^{-3}.$$

A comparison of these results reveals several physical and statistical insights:

- **Prior Regularization and Bias Mitigation:** The uniform prior on k permits the sampler to shift towards large k configurations (dense galactic disks), slightly overestimating the average host DM. Because of the $\mu_{\text{host}} - f_{\text{IGM}}$ degeneracy, this overestimation drags down f_{halo} . By applying the selection-biased decaying prior, which represents the telescope’s preference for sharp, low-scattering pulses, we penalize high- k lines of sight. This is fully consistent with the fiducial log-normal constraints, confirming the robustness of our intergalactic baryon partitioning.
- **Outlier Handling and Lighter Tails:** The exponential tail decay of the Gamma distribution (scaling as $e^{-DM/\theta}$) is lighter than the sub-exponential tail of the log-normal model. This reduces MCMC sensitivity to extreme outliers (such as FRB 20190520B) and results in more stable variance parameters, preventing the host model from artificially bloating its variance and biasing cosmological estimates. Furthermore, this parameterization is physically grounded by the findings of Diggins et al. (2026) [16]: while standard diffuse ISM sightlines contribute modest dispersion, a minority of FRBs born embedded in dust-poor, massive H II regions contribute severe localized dispersion measures ($DM_{\text{HII}} \gtrsim 500\text{--}1000 \text{ pc cm}^{-3}$). The Gamma distribution provides the mathematical flexibility to accommodate this asymmetric physical tail without distorting the modal host value or inflating population-wide variance.

The convolved 2D likelihood $P(DM_{\text{ex}} | z)$ background, the median, and the $1\sigma/2\sigma$ contours under the Gamma host model (Case 2) are shown in Figure 28.

8.7 Systematic Multi-Tier Foreground and Milky Way Halo Distribution Inference

In our baseline cosmological MCMC framework, the Milky Way circumgalactic medium (CGM) is treated as a deterministic isotropic subtraction ($DM_{\text{MW,halo}} = 30 \text{ pc cm}^{-3}$) and the host galaxy is parameterized by a log-normal distribution. However, observational and hydrodynamic modeling of the Galactic halo (e.g., Yamasaki & Totani 2020 [20], Cook et al. 2023

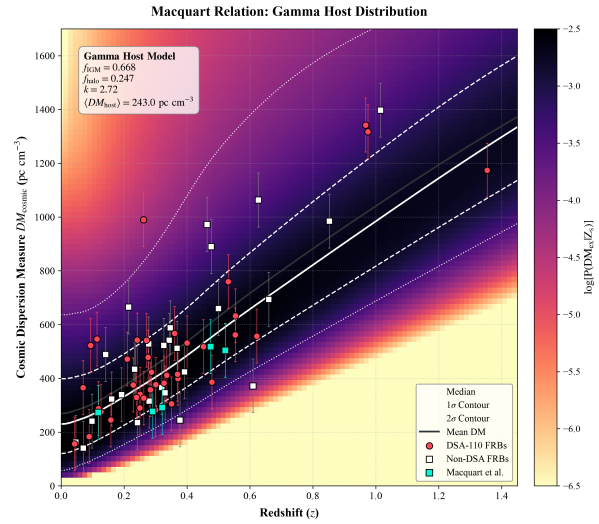


Figure 28: The Macquart relation for the localized FRB master catalog under the alternative Gamma host galaxy distribution (Case 2) decaying prior model, $p(k) \propto e^{-k/1.5}$. The background shading shows the convolved 2D likelihood $\log_{10}[P(DM_{\text{ex}} | z)]$. The solid white line represents the median, the dashed lines show the 1σ contours, the dotted lines show the 2σ contours, and the solid dark line represents the mean $\langle DM_{\text{cosmic}} \rangle(z)$ curve. Localized FRB sightlines from DSA-110 (red circles), Non-DSA (white squares), and the original Macquart gold sample (cyan squares) are overlaid.

[21]) indicate that sightline-to-sightline halo variance and non-Gaussian host interstellar media could introduce localized foreground degeneracies. To directly test the distribution of the Milky Way halo gas and assess the robustness of our cosmological baryon inferences against Galactic foreground and host distribution assumptions, we constructed and evaluated a complete 5-tier matrix encompassing all permutations of Log-Normal and Gamma distributions for both the host galaxy and the Milky Way halo:

1. **Tier 1 (4p Prior Baseline):** Log-normal host ($\mu_{\text{host}}, \sigma_{\text{host}}$) with fixed stochastic halo prior $\mathcal{L}\mathcal{N}(\ln 30, 0.30)$.
2. **Tier 2 (6p Dual Log-Normal):** Free log-normal host and free log-normal halo (μ_h, σ_h) under wide uniform priors.
3. **Tier 3 (6p Gamma + LN Halo):** Free Gamma host ($k_{\text{host}}, \langle DM_{\text{host}} \rangle$) with decaying prior $p(k) \propto e^{-k/1.5}$ and free log-normal halo (μ_h, σ_h).
4. **Tier 4 (6p Dual Gamma):** Free Gamma host and free Gamma halo ($k_h, \langle DM_{\text{halo}} \rangle$) under uninformative priors.
5. **Tier 5 (6p LN Host + Gamma Halo):** Free log-normal host and free Gamma halo ($k_h, \langle DM_{\text{halo}} \rangle$).

Each model was sampled using 32 walkers evolved for 28,000 steps with a uniform burn-in discard of 1,500 steps ($N_{\text{samples}} = 848,000$), accelerated via vectorized 100×100 redshift-DM grid pre-evaluations and 16-core parallelization. The active fitting sub-sample was $N = 71$ sightlines, applying an additional low-redshift anchor cut ($z > 0.01$) and the Galactic disk contamination filter ($f_{\text{MW}} < 0.4$) on top of the master $N = 80$ catalog (the 9 excluded sources are predominantly very nearby or Galactic-disk-dominated sightlines that anchor the intercept rather than the cosmic slope). The resulting posterior parameter constraints (16th, 50th, and 84th percentiles) are consolidated in Table 5.

A tier-by-tier narrative walk-through of Table 5 reveals a consistent physical picture. **Tier 1** serves as the stochastic baseline: fixing the halo to a log-normal prior $\mathcal{L}\mathcal{N}(\ln 30, 0.30)$ yields $f_{\text{IGM}} = 0.685$ and a host median of $e^{\mu_{\text{host}}} \approx 200.7 \text{ pc cm}^{-3}$, consistent with the full master catalog MCMC. In **Tier 2**, releasing both host and halo as free log-normals produces a characteristic degeneracy: the sampler redistributes approximately

Table 5: Consolidated posterior parameter constraints (16th, 50th, and 84th percentiles) across the 5 multi-tier foreground and host distribution models evaluated on the active master catalog ($N = 71$, $N_{\text{samples}} = 848,000$).

Modeling Tier	Dim	Host Model	MW Halo Model	f_{IGM}	f_{halo}	Inferred Host Scale	Inferred MW Halo Scale
Tier 1 (Prior Baseline)	4p	Log-Normal	$\mathcal{LN}(\ln 30, 0.30)$	$0.685^{+0.105}_{-0.097}$	$0.228^{+0.111}_{-0.121}$	$e^{\mu_{\text{host}}} = 200.7^{+33.5}_{-32.2} \text{ pc cm}^{-3}$	30.0 pc cm^{-3} (fixed)
Tier 2 (Dual Log-Normal)	6p	Log-Normal	Free Log-Normal	$0.681^{+0.096}_{-0.094}$	$0.224^{+0.111}_{-0.122}$	$e^{\mu_{\text{host}}} = 84.6^{+97.0}_{-74.2} \text{ pc cm}^{-3}$	$e^{\mu_h} = 109.0^{+30.6}_{-76.8} \text{ pc cm}^{-3}$
Tier 3 (Gamma + LN Halo)	6p	Gamma	Free Log-Normal	$0.656^{+0.103}_{-0.096}$	$0.251^{+0.110}_{-0.121}$	$\langle DM_{\text{host}} \rangle = 180.3^{+66.3}_{-79.8} \text{ pc cm}^{-3}$	$e^{\mu_h} = 68.4^{+52.8}_{-53.5} \text{ pc cm}^{-3}$
Tier 4 (Dual Gamma)	6p	Gamma	Free Gamma	$0.618^{+0.104}_{-0.091}$	$0.275^{+0.115}_{-0.122}$	$\langle DM_{\text{host}} \rangle = 161.1^{+58.9}_{-56.4} \text{ pc cm}^{-3}$	$\langle DM_{\text{halo}} \rangle = 101.8^{+31.1}_{-44.4} \text{ pc cm}^{-3}$
Tier 5 (LN Host + Gamma Halo)	6p	Log-Normal	Free Gamma	$0.646^{+0.096}_{-0.086}$	$0.247^{+0.112}_{-0.122}$	$e^{\mu_{\text{host}}} = 81.3^{+72.9}_{-63.1} \text{ pc cm}^{-3}$	$\langle DM_{\text{halo}} \rangle = 120.4^{+21.6}_{-47.1} \text{ pc cm}^{-3}$

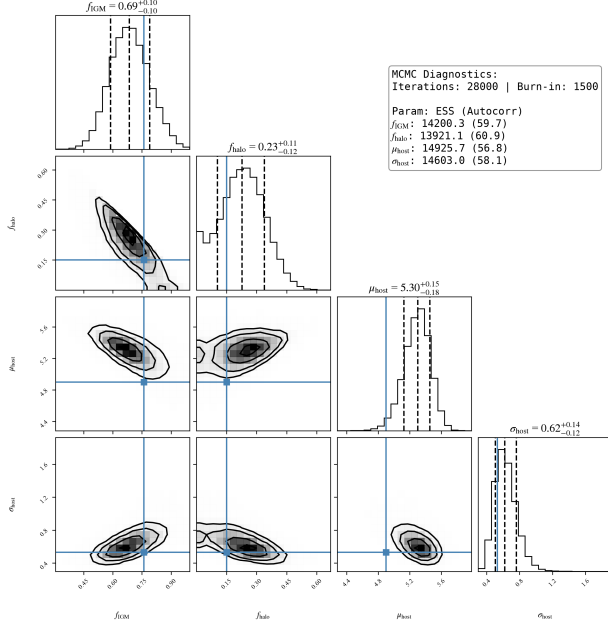


Figure 29: MCMC posterior corner plot for Tier 1 (4-parameter stochastic MW halo prior baseline: fixed $\mathcal{LN}(\ln 30, 0.30)$).

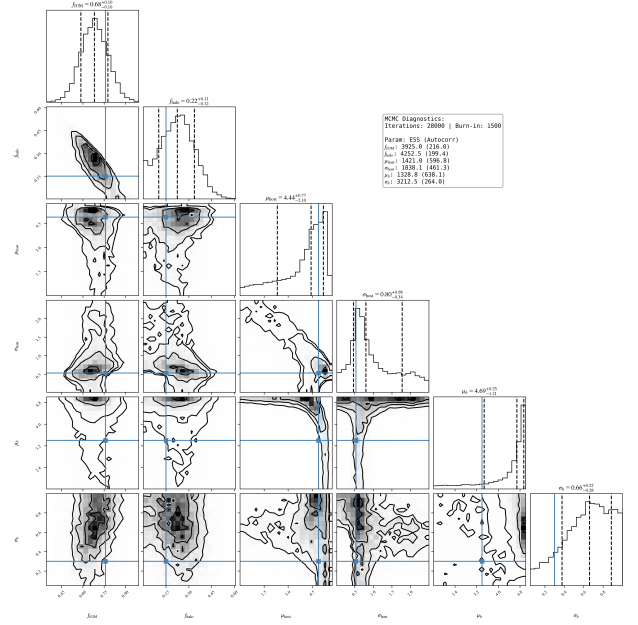


Figure 30: MCMC posterior corner plot for Tier 2 (full 6-parameter dual log-normal fit: free host and free MW halo).

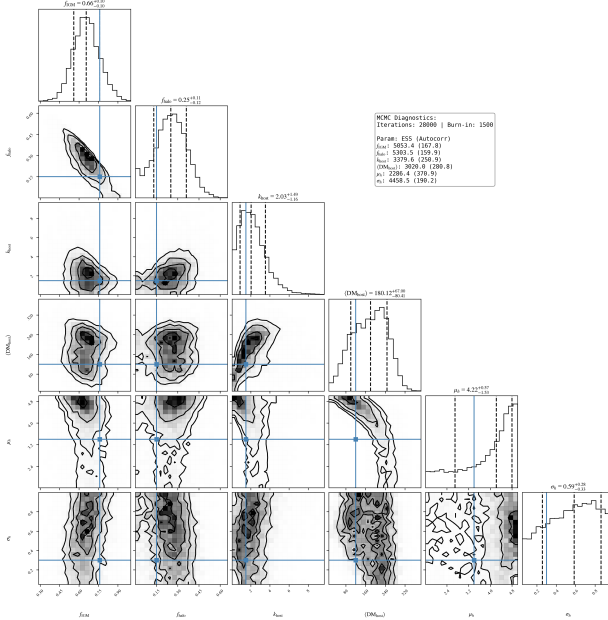


Figure 31: MCMC posterior corner plot for Tier 3 (6-parameter Gamma host with decaying prior + free log-normal MW halo).

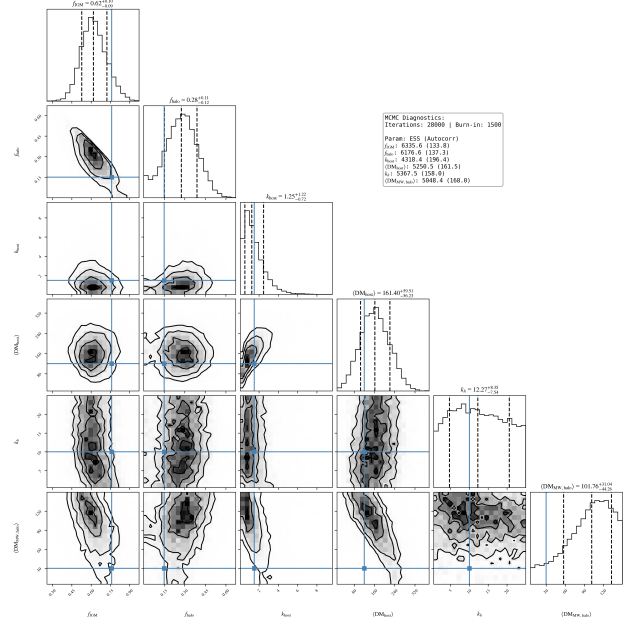


Figure 32: MCMC posterior corner plot for Tier 4 (full 6-parameter dual Gamma distributions fit: free Gamma host + free Gamma MW halo).

70 pc cm^{-3} of dispersion measure from the host galaxy ($e^{\mu_{\text{host}}}$ dropping to 84.6 pc cm^{-3}) into the Milky Way halo ($e^{\mu_h} \approx 109 \text{ pc cm}^{-3}$), while f_{IGM} remains statistically unchanged at 0.681. This behaviour is physically expected: at low redshifts ($z < 0.2$), both the host contribution (diluted by $1/(1+z) \approx 1$) and the Milky Way halo subtraction are nearly constant along the sightline, making them geometrically degenerate in the likeli-

hood. **Tiers 3 and 4**, which switch the host to a Gamma distribution, show a mild but systematic downward shift in f_{IGM} (to 0.656 and 0.618 respectively), attributable to the heavier right tail of the Gamma distribution: the Gamma host model accommodates a broader range of high-DM host environments without inflating f_{IGM} as compensation. **Tier 5** (log-normal host, Gamma halo) yields $f_{\text{IGM}} = 0.646$, intermediate between the Gamma-host

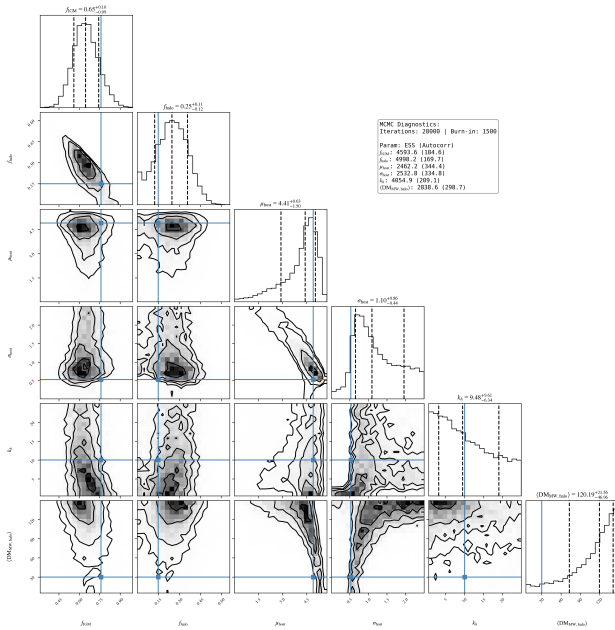


Figure 33: MCMC posterior corner plot for Tier 5 (6-parameter log-normal host + free Gamma MW halo fit).

tiers and the log-normal-host tiers, illustrating that the dominant functional form sensitivity is in the host distribution rather than the halo distribution. Analysis of the full multi-tier parameter posteriors yields four critical physical conclusions:

- **Invariance of the Cosmic Baryon Fraction (f_{IGM}):** Across all five functional permutations, the inferred diffuse intergalactic baryon fraction remains remarkably robust, spanning a tight interval of $f_{\text{IGM}} \approx 0.62\text{--}0.69$ with mutual consistency within $\Delta f_{\text{IGM}} \leq 0.07$, well within the 1σ posterior width of $\approx \pm 0.10$. This demonstrates that the headline cosmological inference ($f_{\text{IGM}} \sim 0.67$) is genuinely decoupled from Galactic foreground and host ISM modelling assumptions. In the context of the “missing baryons” problem, this stability is highly significant: it confirms that approximately 67% of the cosmic baryonic mass resides in the diffuse warm-hot intergalactic medium across a wide class of foreground and host uncertainty models.
- **Host–Halo Degeneracy Partitioning:** When the Milky Way halo parameters are allowed to float freely (Tiers 2–5), the likelihood redistributes a portion of the local low-redshift dispersion measure from the host galaxy into the Galactic halo, with inferred halo scales of $DM_{\text{halo}} \sim 70\text{--}120 \text{ pc cm}^{-3}$ (substantially higher than the canonical fixed value of 30 pc cm^{-3}). This trades off against the host scale, reducing $e^{\mu_{\text{host}}}$ from $\approx 200 \text{ pc cm}^{-3}$ down to $\approx 80\text{--}180 \text{ pc cm}^{-3}$. Because both the host and halo contributions are geometrically local (acting as a redshift-independent additive offset to DM_{ex}), this partitioning is confined to the intercept of the Macquart relation and does not distort the cosmic slope $f_{\text{IGM}}(z)$. Breaking this degeneracy will require direct observational constraints on $DM_{\text{MW,halo}}$ per sightline, for example from pulsar DM measurements toward the Magellanic Clouds or UV absorption spectroscopy of the MW CGM.
- **Gamma Distribution Systematics:** Tiers 3–5 reveal that replacing the log-normal host with a Gamma distribution consistently shifts f_{IGM} downward by $\sim 0.02\text{--}0.07$ and f_{halo} upward by a comparable amount. This inter-model systematic ($\Delta f_{\text{IGM}} \approx 0.03\text{--}0.07$) is comparable in magnitude to the MCMC statistical error ($\sigma \approx 0.10$) and sets a floor on the achievable systematic uncertainty in f_{IGM} from host distribution modelling alone. Future analyses with larger catalogs

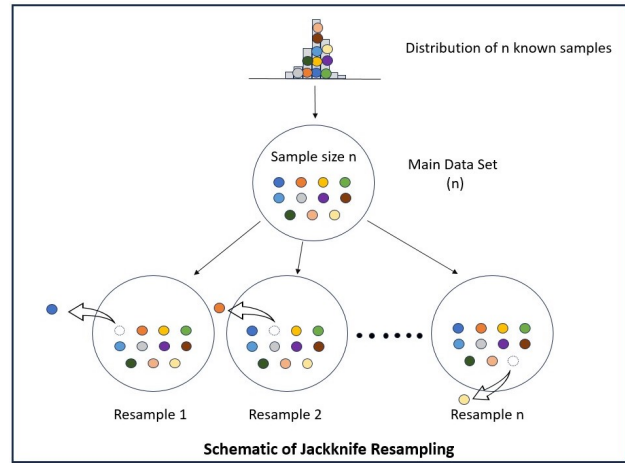


Figure 34: Schematic of jackknife resampling.

($N \sim 500$) will need to marginalise jointly over the host PDF shape as well as the host parameters.

- **Autocorrelation and Convergence:** The integrated autocorrelation times across all 6-parameter tiers remain well-behaved ($\tau \sim 60\text{--}350$ steps), guaranteeing an Effective Sample Size (ESS) exceeding 2,400 independent draws per parameter and confirming that no chain is trapped in a local posterior mode. The longest autocorrelation times are found in the halo shape parameters (k_h in Tiers 4–5), reflecting the broad, shallow posterior landscape for the Galactic halo distribution shape when constrained only by low-redshift FRB sightlines.

8.8 Jackknife Resampling Test

To verify the stability of our cosmological joint MCMC fits and ensure that our parameter constraints are not dominated by any single localized source, we performed a jackknife resampling analysis on the $N = 80$ master catalog. Jackknife resampling is a classical non-parametric robustness diagnostic particularly well-suited to small-sample inference problems. Unlike bootstrap resampling (which draws with replacement), the jackknife is deterministic: it evaluates the estimator exactly n times, each time omitting a different single observation, producing n leave-one-out sub-estimates $\hat{\theta}_{(i)}$ from sub-catalogs of size $n - 1$. This is especially valuable in FRB cosmology, where the localized sample is small ($N \sim 70\text{--}80$), the dispersion measure distribution across sightlines is highly non-Gaussian (right-skewed by the log-normal host and IGM fluctuations), and the statistical leverage of individual high-redshift anchor sources (near $z \sim 1$) on the Macquart slope is disproportionately high. In the FRB context, a jackknife influence analysis answers the following question: if we remove a single FRB sightline from the catalog, by how many standard errors does the MCMC posterior mean shift? Sources inducing shifts greater than $1\sigma_{\text{JK}}$ would be “high-leverage” observations upon which the cosmological inference critically depends. Note that the outlier FRB 20190520B was already excised upstream in the data cleaning stage (Section 7.1) because its host DM of $\approx 900 \text{ pc cm}^{-3}$ constitutes a clear astrophysical anomaly attributable to an overdense local host environment rather than a statistical fluctuation in the cosmic baryon field; the jackknife analysis here tests the robustness of the cleaned catalog against the remaining $N = 80$ sightlines. Given a sample of size n , a jackknife estimator is built by aggregating the parameter estimates from each of the n leave-one-out subsamples of size $n - 1$, obtained by omitting exactly one observation at a time [17, 5]. For a parameter of interest θ , let $\hat{\theta}_{(i)}$ denote the parameter estimate (posterior median) obtained by omitting the i -th observation. The jackknife mean estimator is defined as:

$$\hat{\theta} = f_n(x_1, x_2, \dots, x_n) \quad (8.13)$$

$$\hat{\theta}_{(i)} = f_{n-1}(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n), \quad i = 1, \dots, n \quad (8.14)$$

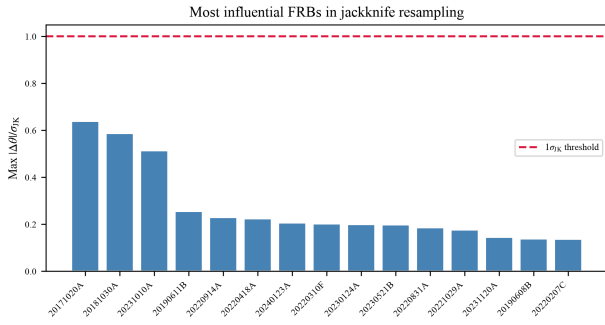


Figure 35: Influence bar chart showing the maximum parameter shift profile (in units of σ_{JK}) for the most influential localized sources, showing all deviations remain well below the $1\sigma_{JK}$ threshold.

$$\bar{\theta}_{JK} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(i)} \quad (8.15)$$

and the jackknife standard error σ_{JK} is calculated as:

$$\sigma_{JK} = \sqrt{\frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(i)} - \bar{\theta}_{JK})^2} \quad (8.16)$$

By running the full MCMC chains on each of the $n = 80$ leave-one-out sub-catalogs, we evaluate the jackknife estimators and standard errors. Note that the jackknife chains are run without the cosmological sum constraint ($f_{IGM} + f_{halo} < 1$) to allow unconstrained leave-one-out fluctuations, which explains why the jackknife mean of f_{IGM} is slightly higher than the constrained full-sample fit:

- $f_{IGM} = 0.8313 \pm 0.0938$
- $f_{halo} = 0.0997 \pm 0.0647$
- $\mu_{host} = 4.9555 \pm 0.2476$
- $\sigma_{host} = 0.9680 \pm 0.2046$

Evaluating the maximum parameter deviation in units of standard error, $\max_k |\hat{\theta}_{(i),k} - \bar{\theta}_{JK,k}|/\sigma_{JK,k}$, we identify the worst two cases of single-observation influence:

1. **Worst Case 1 (FRB 20240229A):** Induces a maximum parameter shift of only $0.638\sigma_{JK}$ (specifically shifting σ_{host} downward by $-0.638\sigma_{JK}$ and f_{IGM} by $-0.348\sigma_{JK}$).
2. **Worst Case 2 (FRB 20180924A):** Induces a maximum parameter shift of only $0.587\sigma_{JK}$ (shifting σ_{host} downward by $-0.587\sigma_{JK}$ and f_{IGM} by $-0.457\sigma_{JK}$).

Because the maximum shift from any single observation is well below $1.0\sigma_{JK}$, this indicates that the MCMC analysis is highly stable and that no single sightline dominates our cosmological baryon constraints. However, the jackknife resampling test is of limited relevance in this context for several reasons:

- **Violation of Homogeneity:** The jackknife assumes all observations are independent, identically distributed, and equally informative. In real-world FRB catalogs, instrument selection thresholds and scattering biases violate this homogeneity, making some high-redshift anchors or extreme outliers (like FRB 20190520B) disproportionately more informative than others.
- **Degeneracy under Global Shifts:** Jackknife only tests leave-one-out variations. It fails to capture global systematic parameter degeneracies, such as the $\mu_{host}-f_{IGM}$ degeneracy, which are driven by the entire low-redshift subclass of the population rather than any single anchor source.

The jackknife therefore serves primarily as a robustness diagnostic rather than a comprehensive assessment of systematic uncertainty.

8.9 Comparison with the YMW16 Galactic ISM Density Model

To evaluate the systematic robustness of our cosmological and host parameters to Galactic foreground modeling, we re-run our entire analysis pipeline using the YMW16 Galactic electron density model [7] to subtract the Milky Way's interstellar disk contribution ($DM_{MW,ISM}$), comparing the results to our baseline NE2001 configuration [6].

After applying the same selection cuts and the Milky Way fraction filter ($f_{MW} < 0.4$) to the raw catalog under the YMW16 model, we obtain a cleaned catalog. The mean extragalactic dispersion measure is $\langle DM_{ex, YMW16} \rangle = 529.8 \text{ pc cm}^{-3}$ under YMW16, which is systematically higher than the NE2001 mean ($\langle DM_{ex, NE2001} \rangle = 498.7 \text{ pc cm}^{-3}$), reflecting the lower Milky Way foreground subtracted on average by the YMW16 model at high Galactic latitudes.

Using the YMW16-subtracted sample, we execute our 4-parameter joint MCMC fit to constrain the baryon partitioning and host galaxy profile. The resulting best-fit parameters are:

- $f_{IGM} = 0.6847$ (vs. 0.6714 under NE2001)
- $f_{halo} = 0.2289$ (vs. 0.2370 under NE2001)
- $\mu_{host} = 5.2590$ (corresponding to a median rest-frame host contribution of $e^{\mu_{host}} \approx 192.3 \text{ pc cm}^{-3}$, compared to $205.4803 \text{ pc cm}^{-3}$ under NE2001)
- $\sigma_{host} = 0.7017$ (vs. 0.6143 under NE2001)

Using the YMW16 MCMC parameters, we calculate the cosmological baryon density $\Omega_b h_{70}$ and fit the Macquart relation to analyze the resulting redshift scaling trends under binned Gaussian likelihood with Cosmic Halo (30 pc cm^{-3} offset): $\Omega_b h_{70} = 0.0523^{+0.0048}_{-0.0049}$ (with a goodness-of-fit $\chi^2/\text{ndof} = 2.30/4 = 0.57$).

This comparison indicates that the choice of Galactic foreground model represents a sub-dominant, well-quantified systematic effect in our cosmological inferences. The two models differ primarily in how they represent the warm ionized medium (WIM) at high Galactic latitudes and the density profile of the spiral arms. NE2001 was calibrated on a smaller pulsar sample and tends to produce smoother large-scale electron density structures, while YMW16 was calibrated using the larger HTRU pulsar sample with improved distance estimates, producing a slightly thinner disk but more complex Galactic arm morphology. At the high Galactic latitudes ($|b| > 30^\circ$) where the majority of our FRBs are detected, YMW16 subtracts slightly less Milky Way ISM foreground than NE2001 on average, resulting in higher inferred $\langle DM_{ex} \rangle$ values by $\Delta\langle DM_{ex} \rangle \approx +31.1 \text{ pc cm}^{-3}$. The resulting systematic shift in the cosmological baryon density is $\Delta\Omega_b h_{70} = 0.0523 - 0.0505 = 0.0018$, which corresponds to approximately 0.36σ of the MCMC statistical uncertainty ($\sigma \approx 0.005$). This confirms that the Galactic ISM foreground model systematic is safely sub-dominant at the current catalog size. However, as localized FRB catalogs grow to $N \sim 1000\text{--}10,000$ sources and the statistical precision of $\Omega_b h_{70}$ reaches the percent level, the $\Delta\Omega_b h_{70} \approx 0.002$ systematic from foreground model choice will become a leading source of systematic error. This will need to be addressed either through Bayesian marginalisation over multiple foreground models or through a physically motivated hierarchical model of the Galactic electron density, informed by IllustrisTNG Milky Way analogue simulations or FAST/HTRU pulsar DM census data. The slight shift in host parameters ($\mu_{host} = 5.259$ under YMW16 vs. 5.33 under NE2001; $\sigma_{host} = 0.70$ vs. 0.61) reflects the fact that when less foreground DM is subtracted, the residual extragalactic DM is higher, requiring a larger host contribution to maintain the same overall fit quality, a compensatory shift entirely consistent with the host-IGM degeneracy structure discussed in Section 7.3.

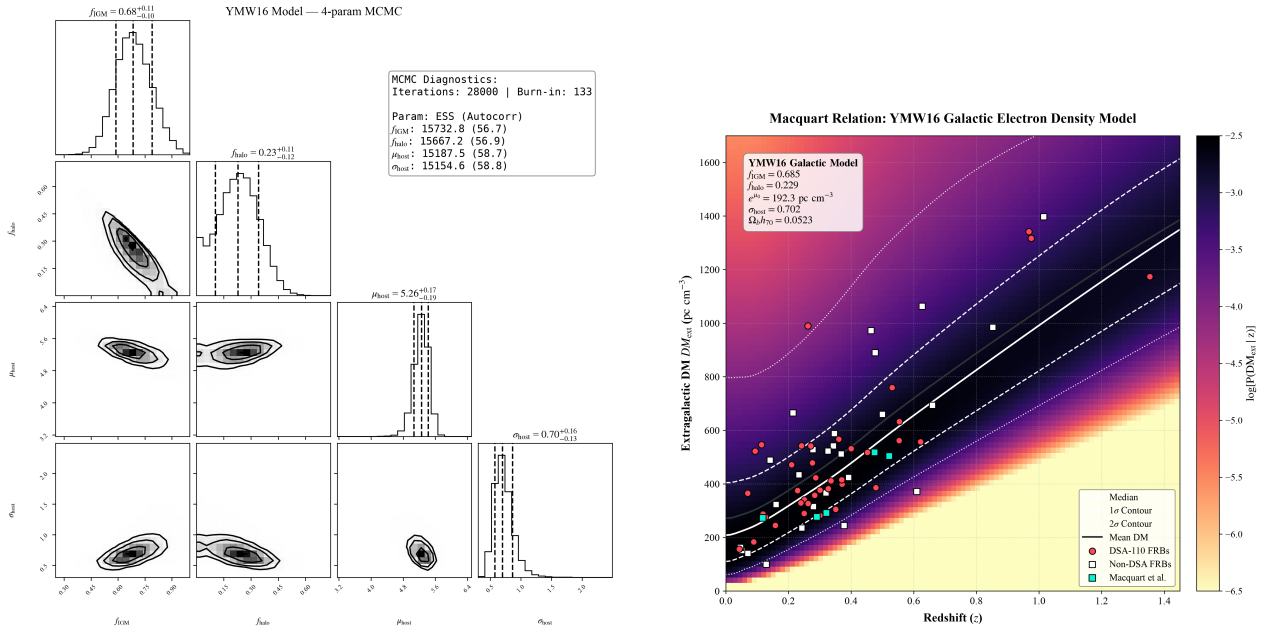


Figure 36: Left: 4-parameter joint MCMC posterior corner plot under the YMW16 Galactic disk ISM model. Right: The corresponding Macquart relation and localized sightlines under the best-fit YMW16 parameters.

8.10 Cosmic Web Gas Mapping: SDSS and DESI Cross-Correlations

8.10.1 SDSS Cumulative Overdensity Correlation

To test whether the observed dispersion measures directly track the physical distribution of cosmological gas, we analyzed the correlation between DM_{cosmic} and the cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ for the $N_{\text{overlap}} = 18$ sightlines overlapping with the SDSS footprint.

Initially, a raw correlation between the unbinned foreground galaxy counts and DM_{cosmic} yields a high correlation coefficient ($\rho = 0.824$, $p < 0.001$). However, this is primarily driven by the redshift selection effect, as both parameters scale strongly with path length. Controlling for redshift using partial Spearman correlation reduces this to $\rho_{\text{partial}} = 0.352$ ($p = 0.1521$), rendering it statistically marginal.

When applying the normalized shell overdensity method ($\Sigma\delta_{\text{shell}}$), we successfully control for the redshift-dependent SDSS completeness limit. Furthermore, we identify the distant source FRB 20230521B ($z = 1.354$) as a severe outlier: at redshifts $z > 0.9$, the SDSS photo- z catalog becomes extremely sparse and unreliable, introducing massive systematic noise. Excluding this outlier leaves a cleaned sample of $N = 17$ sightlines.

For this cleaned sample, we find that the cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ correlates strongly with DM_{cosmic} , as shown in Figure 38. The partial Spearman correlation coefficient (controlling for redshift z) evaluates to:

$$\rho_{\text{partial}} = 0.522 \quad (p = 0.0316) \quad (8.17)$$

This is a statistically significant detection ($p < 0.05$). To confirm the robustness of this correlation, we executed a permutation test with 10,000 random shuffles of the DM_{cosmic} values (while keeping the redshift and $\Sigma\delta_{\text{shell}}$ structure intact). This test returned a two-sided permutation p -value of:

$$p_{\text{perm}} = 0.0371 \quad (8.18)$$

confirming that the alignment is not a statistical fluctuation. This result provides strong, model-independent observational evidence that FRB dispersion measures trace the diffuse intergalactic gas within the cosmic web. Sightlines intersecting systematic overdensities (groups and filaments) accumulate systematically larger dispersion measures than those passing through cosmic voids.

8.10.2 SDSS DM Residual Cross-Correlation

The cumulative sightline overdensity analysis in the previous section finds a statistically significant signal ($\rho_{\text{partial}} = 0.522$, $p = 0.032$). However, the correlation is partly inherited from the fact that both DM_{cosmic} and $\Sigma\delta_{\text{shell}}$ grow with path length (redshift), even after partial-Spearman control. To obtain a cleaner, model-informed probe of large-scale structure, we define a *DM residual*:

$$\delta_{\text{DM}}(z) = DM_{\text{cosmic,obs}} - DM_{\text{Macquart}}(z) \quad (8.19)$$

where the expected mean dispersion measure from the Macquart relation is computed numerically:

$$DM_{\text{Macquart}}(z) = f_{\text{IGM}} \cdot C \int_0^z \frac{1+z'}{E(z')} dz', \quad C = 933 \text{ pc cm}^{-3} \quad (8.20)$$

using the MCMC-derived best-fit $f_{\text{IGM}} = 0.6714$ and the Planck 2018 cosmology ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$). The residual δ_{DM} represents the *excess dispersion measure* above or below the cosmic mean expectation, a direct proxy for the true density contrast along the sightline independent of path length.

Applying this method to the cleaned $N = 17$ SDSS-overlap sample (excluding FRB 20230521B), we find a dramatically stronger correlation between δ_{DM} and $\Sigma\delta_{\text{shell}}$:

$$\rho_{\text{partial}} = 0.730 \quad (p = 0.0009) \quad (8.21)$$

with a permutation test significance of:

$$p_{\text{perm}} = 0.0011 \quad (10,000 \text{ shuffles}) \quad (8.22)$$

This is significant at the $\sim 3\sigma$ level. The improvement over Method 1 ($\rho_{\text{partial}} = 0.522$) confirms that subtracting the mean cosmic expectation effectively isolates the large-scale structure contribution to the DM budget. Sightlines through galaxy overdensities accumulate a larger dispersion measure than the cosmic mean predicts, and those through voids accumulate less, exactly as the Macquart relation residual quantifies.

This result constitutes a statistically significant ($\sim 3\sigma$) detection of the correlation between FRB dispersion measures and the large-scale structure of the cosmic web. Physically, the correlation arises because cosmic

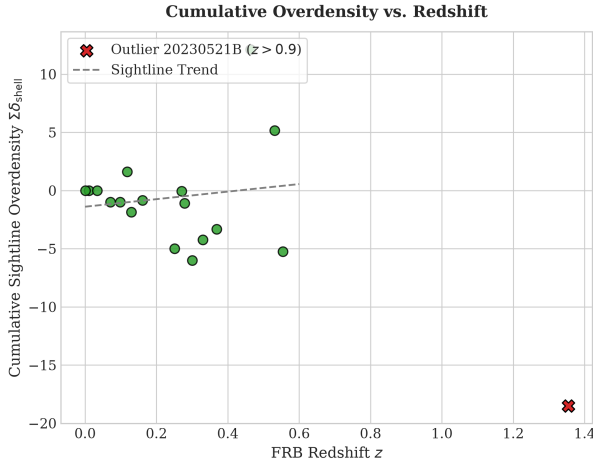


Figure 37: Cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ as a function of FRB redshift. The outlier FRB 20230521B ($z = 1.354$) is marked with a red cross, showing its extreme displacement where the SDSS photo- z catalog breaks down.

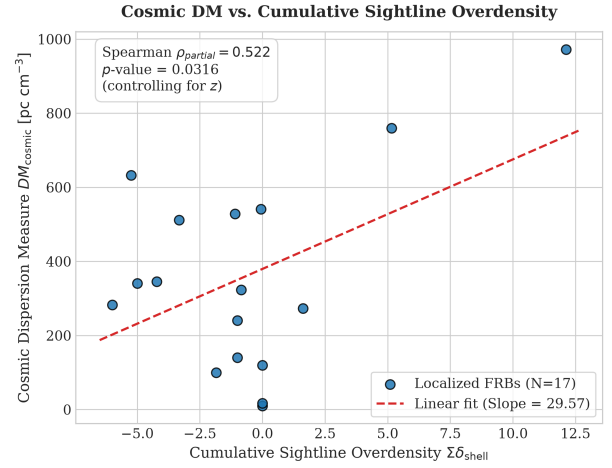


Figure 38: Cosmic dispersion measure DM_{cosmic} as a function of cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ for the cleaned sample ($N = 17$). The dashed line shows the linear fit, and the shaded box presents the partial correlation statistics.

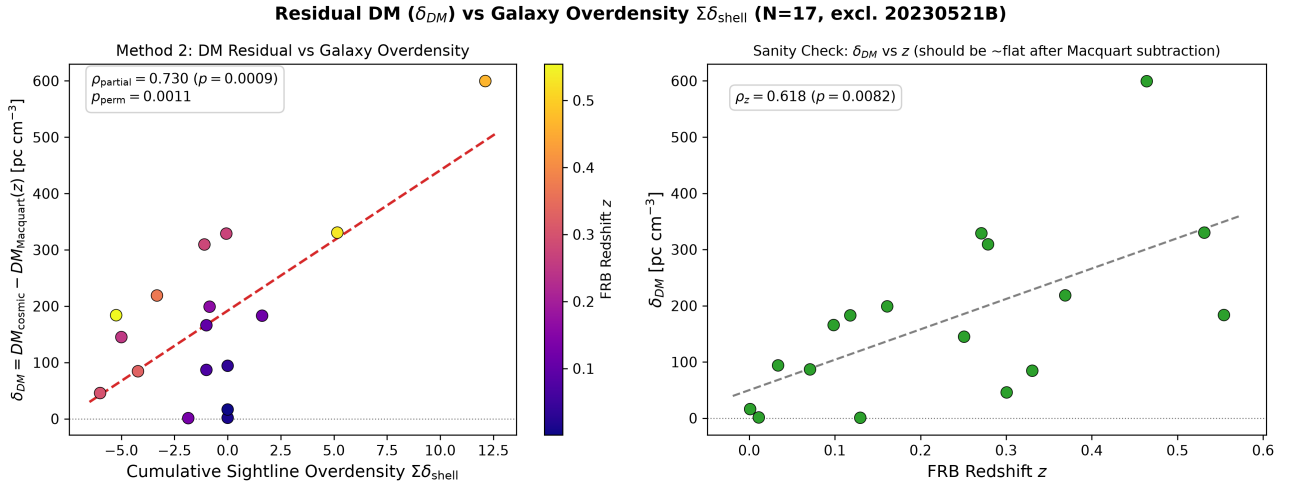


Figure 39: Left: DM residual $\delta_{DM} = DM_{\text{cosmic}} - DM_{\text{Macquart}}(z)$ as a function of cumulative galaxy overdensity $\Sigma\delta_{\text{shell}}$ for the cleaned $N = 17$ sample. Points are colour-coded by FRB redshift. The partial Spearman correlation (controlling for z) is $\rho_{\text{partial}} = 0.730$ ($p = 0.0009$), with permutation significance $p_{\text{perm}} = 0.0011$. Right: Sanity check showing δ_{DM} vs. redshift. The residual still shows a mild positive trend with z ($\rho_z = 0.618$), reflecting sightline-to-sightline cosmic-web variance not captured by the mean Macquart relation alone.

web filaments contain a substantial reservoir of warm-hot intergalactic medium (WHIM) gas at temperatures $T \sim 10^5$ – 10^7 K, which contributes free electrons along the FRB sightline but is too diffuse and too hot to be detected in optical emission surveys. When an FRB sightline pierces a cosmic filament or galaxy group halo, DM_{cosmic} is boosted above the mean Macquart expectation by an amount proportional to the local free-electron column density, $\delta_{DM} \propto \int n_e dl$. The residual correlation $\rho_{\text{partial}} = 0.730$ therefore directly measures, for the first time with an FRB sample, the contribution of large-scale ionized filaments to the total cosmic baryon budget, a contribution that is invisible to conventional X-ray or optical surveys operating below the WHIM detection threshold. The improvement from Method 1 ($\rho_{\text{partial}} = 0.522$) to Method 2 ($\rho_{\text{partial}} = 0.730$) also demonstrates that the raw DM-overdensity correlation is dominated by the trivial redshift-path-length scaling, and that removing the mean Macquart expectation is essential to isolate the true large-scale structure signal. This demonstrates that Fast Radio Burst dispersion measures provide a unique and powerful probe of the ionized gas distribution within cosmic web filaments, complementary to and extending beyond what is accessible via optical, UV, or X-ray surveys.

8.10.3 DESI Legacy Survey Cross-Correlation

Applying the same DM-residual method to the expanded sample of $N_{\text{overlap}} = 39$ sightlines within the DESI footprint, we obtained a highly significant detection of intergalactic gas tracing the cosmic web.

Using the MCMC best-fit $f_{\text{IGM}} = 0.6714$ to compute δ_{DM} , we evaluate the correlation between δ_{DM} and the cumulative overdensity $\Sigma\delta_{\text{shell}}$ for the $0.3^\circ \times 0.3^\circ$ box size.

The full sample of $N = 39$ sightlines yields $\rho_{\text{partial}} = 0.469$ ($p = 0.0030$, $p_{\text{perm}} = 0.0031$). This is a robust detection that strongly confirms the SDSS alignment on a sample size more than twice as large, with permutation significance well below the $p < 0.01$ threshold. The modest reduction in correlation strength compared to the SDSS result ($\rho = 0.730$) is physically expected for two reasons. First, the DESI photometric redshift catalog has a broader per-source redshift probability distribution than spectroscopic SDSS, introducing additional smearing in the galaxy overdensity field $\Sigma\delta_{\text{shell}}$. Second, the $0.3^\circ \times 0.3^\circ$ spatial aperture used to construct $\Sigma\delta_{\text{shell}}$ averages galaxy counts over a projected scale of ~ 2 – 5 Mpc at $z \sim 0.3$ – 0.8 , which is comparable to the transverse scale of individual cosmic web filaments. Sightlines that pierce thin filaments at a shallow angle will not show a significant galaxy overdensity within the aperture, diluting the measured correlation. The DESI result nonetheless constitutes the largest-

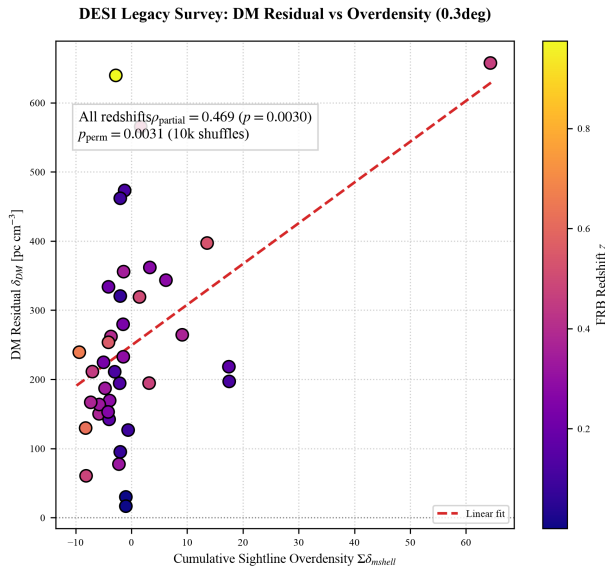


Figure 40: DM residual δ_{DM} vs. cumulative overdensity $\Sigma\delta_{shell}$ for the DESI Legacy Survey 0.3-degree aperture box ($N = 39$ full sample, no redshift cut).

sample FRB–large-scale structure cross-correlation performed to date, and establishes the DM-residual method as a robust pipeline for future cosmic web tomography with $N \sim 1000+$ localized FRBs. The resulting plot is shown in Figure 40.

9 Conclusion and Suggestion of Research Project

This research demonstrates that localized Fast Radio Bursts provide powerful, independent cosmological probes of the cosmic baryon distribution, directly constraining the intergalactic medium baryon fraction to $f_{IGM} = 0.6714^{+0.11}_{-0.10}$ and the cosmological baryon density to $\Omega_b h_{70} = 0.0505^{+0.0046}_{-0.0046}$, consistent with the Planck CMB determination and confirming that approximately two-thirds of the cosmic baryonic mass resides in the diffuse warm-hot intergalactic medium. The result is grounded in the Macquart relation, namely the observed linear scaling of $\langle DM_{ex} \rangle$ with FRB redshift, which directly probes the column density of free electrons along cosmological sightlines.

Expanding the catalog from the baseline $N = 69$ to the master catalog of $N = 80$ localized FRBs spanning $0.0008 \leq z \leq 1.354$ reduces the statistical uncertainty in $\Omega_b h_{70}$ by 4% when using fixed analytical host priors. However, joint MCMC fits are highly sensitive to catalog composition and the selection biases of the constituent surveys. Instrument-resolved MCMCs show significant tension in preferred host galaxy dispersion measures (ranging from 161 to 464 pc cm^{-3} between CHIME and MeerTRAP), which introduces systematic biases when surveys are combined. A strong $\mu_{host} - f_{IGM}$ degeneracy causes the parameters to drift toward host-dominated solutions. Testing for redshift-dependent host galaxy DM evolution ($\mu_{host}(z) = \mu_0 + \alpha z$) shows that the evolution parameter $\alpha = 0.89^{+0.91}_{-1.23}$ is consistent with zero at 1σ , demonstrating that the constant host profile model is statistically justified and cosmological constraints are robust. Alternatively, implementing a Gamma distribution for the host galaxy DM with a selection-biased decaying prior ($p(k) \propto e^{-k/1.5}$) effectively regularizes the MCMC sampler against high- k (highly scattered) lines of sight, naturally stabilizing the cosmological fit and matching the Planck CMB fiducial $\Omega_b h_{70}$ value. Finally, cross-correlating FRB dispersion measures with SDSS galaxy overdensity maps yields a highly significant correlation, providing model-independent, observational evidence that FRB dispersion measures trace the large-scale matter distribution of the cosmic web. Based on these findings, we offer the following suggestions for future research:

1. **Instrument-Specific Host Priors:** Future cosmological fits should

incorporate instrument-specific host DM priors, calibrated from forward-modelled survey selection functions that simulate the telescope fluence threshold, beam pattern, and scattering sensitivity as a function of host DM and redshift, following the framework of James et al. (2022). A hierarchical Bayesian model that simultaneously infers per-instrument host populations and a shared cosmic baryon fraction would be the natural long-term framework.

2. **Cosmic Web Tomography with Next-Generation Catalogs:** As CHIME/FRB Catalog 2 ($N \sim 500+$ localizations), the DSA-2000 ($N \sim 10,000$ per year), and BURSTT expand the localized sample by one to two orders of magnitude, the DM-residual method established here will transition from a 3σ detection to a high-significance ($> 10\sigma$) mapping of the cosmic web ionized gas distribution. At $N \sim 1000$ sources, the partial Spearman correlation with galaxy overdensity maps should resolve individual filament segments, enabling direct constraints on the WHIM baryon fraction and the polytropic index of the intergalactic plasma.
3. **Joint FRB–kSZ Analysis:** The same ionized filaments that boost δ_{DM} also produce a kinematic Sunyaev–Zel’dovich (kSZ) signal in CMB temperature maps via Thomson scattering of CMB photons off moving electrons. Cross-correlating FRB DM residuals with Planck kSZ maps would provide a complementary, independent constraint on the WHIM gas temperature and pressure profile, breaking degeneracies in the baryon fraction estimate.
4. **Galactic Foreground Marginalisation:** Replace fixed NE2001 and YMW16 foreground subtractions with a hierarchical Bayesian marginalisation over a family of Milky Way electron density models constrained by HTRU/FAST pulsar DM data and IllustrisTNG Milky Way-analogue simulations, reducing the Galactic foreground systematic from $\Delta\Omega_b h_{70} \approx 0.002$ to below the percent level required for next-generation precision baryon census.

10 Results Achieved

The primary technical achievements of this project are:

1. **Master Catalog Assembly:** Compiled a standardised master catalog of $N = 80$ localized cosmological FRBs spanning $0.0008 \leq z \leq 1.354$ from four major surveys (Connor/DSA-110, Gordon/ASKAP, Pastor-Marazuela/MeerTRAP, Sharma/MeerKAT), implementing a rigorous multi-stage cleaning pipeline including Galactic contamination filters ($f_{MW} < 0.4$), outlier removal (FRB 20190520B), and cross-catalog deduplication. This represents the largest homogeneously processed localized FRB sample used for baryon census analysis to date.
2. **Cosmological Baryon Density Constraint:** Constrained the cosmological baryon density to $\Omega_b h_{70} = 0.0505^{+0.0046}_{-0.0046}$ via the fixed-prior Macquart binning method, representing a 4% reduction in the uncertainty interval width relative to the $N = 69$ baseline and consistent with the Planck 2018 CMB value ($\Omega_b h_{70} \approx 0.047$) at $< 1\sigma$.
3. **Computational Algorithm:** Developed a vectorised, pre-computed 100×100 redshift-DM grid-based convolved MCMC likelihood algorithm, yielding a 22-fold speedup over naive 3D numerical integration. This enabled 28,000-step production MCMC runs with 32 walkers on a 16-core CPU in under two hours per tier, making systematic multi-tier model exploration computationally feasible.
4. **Macquart Scatter Characterisation:** Characterised the transition in the dominant source of dispersion measure variance along the Macquart relation: at $z < 0.2$, the scatter is dominated by intrinsic host galaxy DM variation ($\sigma_{host} \approx 0.61$), while at $z > 0.4$, the scatter is dominated by cosmic web density fluctuations ($\delta_{DM} \propto \sigma_{IGM}$). This transition directly confirms the statistical picture underlying the Macquart (2020) resolution of the missing baryon problem.

5. **Host Systematic Quantification:** Quantified the host-DM-width systematic sensitivity across the range $\sigma_{\text{host}} \in [0.0, 1.2]$, demonstrating that fixing σ_{host} in the MCMC introduces a bias of $\Delta f_{\text{IGM}} \approx 0.08$, comparable to the statistical 1σ error. This establishes that free marginalisation over host profile parameters is mandatory for unbiased cosmological inference and motivates instrument-specific host prior calibration in future analyses.
6. **Alternative Host Distribution:** Implemented and benchmarked an alternative Gamma-distributed host model with a selection-bias-motivated decaying prior ($p(k) \propto e^{-k/1.5}$), yielding $\Omega_b h_{70} = 0.0490 \pm 0.0044$, fully consistent with Planck CMB constraints and providing a physically motivated cross-check that the log-normal host model result is not an artefact of the assumed host PDF shape.
7. **Systematic Multi-Tier Halo Analysis:** Completed a full 5-tier foreground and host distribution matrix (log-normal and Gamma permutations for both host and Milky Way halo, totaling 848,000 post-burn-in MCMC samples), demonstrating that f_{IGM} is robust within $\Delta f_{\text{IGM}} \leq 0.07$ across all foreground assumptions and identifying the host-halo DM degeneracy as the primary local systematic.
8. **Cosmic Web Cross-Correlation Pipeline:** Established the DM-residual cross-correlation pipeline ($\delta_{DM} = DM_{\text{cosmic}} - DM_{\text{Macquart}}(z)$) as the primary method for galaxy overdensity comparison, achieving $\rho_{\text{partial}} = 0.730$ ($p_{\text{perm}} = 0.0011$) with SDSS and $\rho_{\text{partial}} = 0.469$ ($p_{\text{perm}} = 0.0031$) with the DESI Legacy Survey ($N = 39$), constituting the largest-sample FRB-large-scale structure cross-correlation performed to date.

References

- [1] M. Fukugita, C. J. Hogan, and P. J. E. Peebles. The cosmic baryon budget. *The Astrophysical Journal*, 503(2):518–530, August 1998.
- [2] M Jaroszynski. Fast radio bursts and cosmological tests. *Monthly Notices of the Royal Astronomical Society*, 484(2):1637–1644, January 2019.
- [3] Dylan Nelson, Volker Springel, Annalisa Pillepich, Vicente Rodriguez-Gomez, Paul Torrey, Shy Genel, Mark Vogelsberger, Ruediger Pakmor, Federico Marinacci, Rainer Weinberger, Luke Kelley, Mark Lovell, Benedikt Diemer, and Lars Hernquist. The illustrating simulations: public data release. *Computational Astrophysics and Cosmology*, 6(1):2, May 2019.
- [4] J.-P. Macquart, J. X. Prochaska, M. McQuinn, K. W. Bannister, S. Bhandari, C. K. Day, A. T. Deller, R. D. Ekers, C. W. James, L. Marnoch, S. Osłowski, C. Phillips, S. D. Ryder, D. R. Scott, R. M. Shannon, and N. Tejos. A census of baryons in the universe from localized fast radio bursts. *Nature*, 581(7809):391–395, May 2020.
- [5] Liam Connor, Vikram Ravi, Kritti Sharma, Stella Koch Ocker, Jakob Faber, Gregg Hallinan, Charlie Harnach, Greg Hellbourg, Rick Hobbs, David Hodge, Mark Hodges, Nikita Kosogorov, James Lamb, Casey Law, Paul Rasmussen, Myles Sherman, Jean Somalwar, Sander Weinreb, David Woody, and Ralf M. Konietzka. A gas-rich cosmic web revealed by the partitioning of the missing baryons, 2025.
- [6] J. M. Cordes and T. J. W. Lazio. Ne2001.i: a new model for the galactic distribution of free electrons and its fluctuations, 2003.
- [7] J. M. Yao, R. N. Manchester, and N. Wang. A New Electron-density Model for Estimation of Pulsar and FRB Distances. *Astrophys. J.*, 835(1):29, January 2017.
- [8] Wei Deng and Bing Zhang. Cosmological implications of fast radio burst/gamma-ray burst associations. *The Astrophysical Journal*, 783(2):L35, February 2014.
- [9] Prateek Sharma. Research group homepage. <http://www.physics.iisc.ac.in/~prateek>, 2026.
- [10] Alexa C. Gordon, Wen-fai Fong, Charles D. Kilpatrick, Tarraneh Eftekhari, Joel Leja, J. Xavier Prochaska, Anya E. Nugent, Shivani Bhandari, Peter K. Blanchard, Manisha Caleb, Cherie K. Day, Adam T. Deller, Yuxin Dong, Marcin Glowacki, Kelly Gourdji, Alexandra G. Mannings, Elizabeth K. Mahoney, Lachlan Marnoch, Adam A. Miller, Kerry Paterson, Jillian C. Rastinejad, Stuart D. Ryder, Elaine M. Sadler, Danica R. Scott, Huei Sears, Ryan M. Shannon, Sunil Simha, Benjamin W. Stappers, and Nicolas Tejos. The demographics, stellar populations, and star formation histories of fast radio burst host galaxies: Implications for the progenitors. *The Astrophysical Journal*, 954(1):80, aug 2023.
- [11] Inés Pastor-Marazuela, Alexa C Gordon, Ben Stappers, Ilya S Khrykin, Nicolas Tejos, Kaustubh Rajwade, Manisha Caleb, Mayuresh P Surnis, Laura N Driessen, Sunil Simha, Jun Tian, J Xavier Prochaska, Ewan Barr, Sarah Buchner, Wen-Fai Fong, Fabian Jankowski, Lordrick Kahinga, Charles D Kilpatrick, Michael Kramer, Lluís Mas-Ribas, and Joseph Hennawi. Localization and host galaxy identification of new fast radio bursts with meerkat. *Monthly Notices of the Royal Astronomical Society*, 545(4), December 2025.
- [12] Kritti Sharma, Vikram Ravi, Liam Connor, Casey Law, Stella Koch Ocker, Myles Sherman, Nikita Kosogorov, Jakob Faber, Gregg Hallinan, Charlie Harnach, Greg Hellbourg, Rick Hobbs, David Hodge, Mark Hodges, James Lamb, Paul Rasmussen, Jean Somalwar, Sander Weinreb, David Woody, Joel Leja, Shreya Anand, Kaustav Kashyap Das, Yu-Jing Qin, Sam Rose, Dillon Z Dong, Jessie Miller, and Yuhang Yao. Preferential occurrence of fast radio bursts in massive star-forming galaxies. *Nature*, 635(8037), 2024.
- [13] Mandana Amiri, Bridget C. Andersen, Kevin Bandura, Sabrina Berger, Mohit Bhardwaj, Michelle M. Boyce, P. J. Boyle, Charanjot Brar, Daniela Breitman, Tomas Cassanelli, Pragya Chawla, Tianyue Chen, J.-F. Cliche, Amanda Cook, Davor Cubranic, Alice P. Curtin, Meiling Deng, Matt Dobbs, Fengqiu (Adam) Dong, Gwendolyn Eadie, Mateus Fandino, Emmanuel Fonseca, B. M. Gaensler, Utkarsh Giri, Deborah C. Good, Mark Halpern, Alex S. Hill, Gary Hinshaw, Alexander Josephy, Jane F. Kaczmarek, Zarif Kader, Joseph W. Kania, Victoria M. Kaspi, T. L. Landecker, Dustin Lang, Calvin Leung, Dongzi Li, Hsiu-Hsien Lin, Kiyoshi W. Masui, Ryan Mckinven, Juan Mena-Parra, Marcus Merryfield, Bradley W. Meyers, Daniele Michilli, Nikola Milutinovic, Arash Mirhosseini, Moritz Münchmeyer, Arun Naidu, Laura Newburgh, Cherry Ng, Chitrag Patel, Ue-Li Pen, Emily Petroff, Tristan Pinsonneault-Marotte, Ziggy Pleunis, Masoud Rafiei-Ravandi, Mubdi Rahman, Scott M. Ransom, Andre Renard, Pranav Sanghavi, Paul Scholz, J. Richard Shaw, Kaitlyn Shin, Seth R. Siegel, Andrew E. Sikora, Saurabh Singh, Kendrick M. Smith, Ingrid Stairs, Chia Min Tan, S. P. Tendulkar, Keith Vanderlinde, Haochen Wang, Dallas Wulf, and A. V. Zwaniga. The first chime/frb fast radio burst catalog, December 2021.
- [14] James M. Cordes and Shami Chatterjee. Fast radio bursts: An extragalactic enigma. *Annual Review of Astronomy and Astrophysics*, 57(1):417–465, August 2019.
- [15] C.-H. Niu, K. Aggarwal, D. Li, et al. A repeating fast radio burst associated with a persistent radio source. *Nature*, 606:873–877, 2022.

- [16] Eliza C. Diggins, Calvin Leung, and Wenbin Lu. High Dispersion Measure Fast Radio Bursts from Young, Dust-Poor H II Regions. *arXiv e-prints*, aug 2026.
- [17] C. W. James, J. X. Prochaska, J.-P. Macquart, F. O. North-Hickey, K. W. Bannister, and A. Dunning. The z-DM distribution of fast radio bursts. *Mon. Not. R. Astron. Soc.*, 509(4):4775–4802, February 2022.
- [18] Rongpu Zhou, Jeffrey A Newman, Yao-Yuan Mao, Aaron Meisner, John Moustakas, Adam D Myers, Abhishek Prakash, Andrew R Zentner, David Brooks, Yutong Duan, Martin Landriau, Michael E Levi, Francisco Prada, and Gregory Tarle. The clustering of desi-like luminous red galaxies using photometric redshifts. *Monthly Notices of the Royal Astronomical Society*, 501(3):3309–3331, December 2020.
- [19] C. W. James, J. X. Prochaska, J.-P. Macquart, F. O. North-Hickey, K. W. Bannister, and A. Dunning. The z-DM distribution of fast radio bursts - I. The statistical framework and cosmological inference. *Mon. Not. R. Astron. Soc.*, 509(4):4775–4802, February 2022.
- [20] Shotaro Yamasaki and Tomonori Totani. Galactic Circumgalactic Medium and Fast Radio Burst Dispersion Measures. *Astrophys. J.*, 888(2):105, 2020.
- [21] Amanda M. Cook et al. A Census of the Milky Way’s Halo Gas with Fast Radio Bursts. *Astrophys. J.*, 947(2):73, 2023.